

Radio, Information Poverty, and Electoral Politics in Interwar America

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Abstract

Radio reshaped interwar electoral politics by delivering political information to citizens excluded from print media by literacy barriers. Using AM signal propagation as a geographic instrument for county-level radio ownership (1920–1942), I find radio raised Democratic presidential vote share within states, with the effect strongest in high-illiteracy counties. Vote-count decomposition shows the shift reflects Republican vote loss rather than Democratic gain on average, a pattern consistent with habitual Republican voters switching or withdrawing once radio revealed the gap between inherited partisanship and economic interests. Effects vanish in the Jim Crow South, where one-party dominance left few such voters the shock could move. At the district level, radio shifted legislator ideology leftward in high-illiteracy districts, with both parties responding. These results show that a new information technology's political consequences rest not only on its content, but also on the audiences it reaches and the information environment they already inhabit.

Keywords: radio, media, electoral accountability, political representation, interwar America

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1 Introduction

Before radio, political news in the United States traveled almost exclusively through print. For the millions of adults who could not read, that meant little independent political news beyond what rallies, canvassers, and neighbors carried by word of mouth. Illiteracy exceeded 10% in many states in the 1920s and 1930s, and rural communities often lacked daily newspapers. For these voters the cost of sustained, unmediated political information — which candidates were running, how their representatives had voted — was effectively prohibitive. The information environment was structured around print, and print required literacy.

Commercial radio broadcasting, emerging in 1920, changed this. Radio offered political news as spoken word, accessible regardless of literacy, and within two decades ownership rose from essentially zero to nearly 70% of American households. For voters excluded from print media, radio was not merely a new channel but the first.

Democratic accountability depends on voters' ability to monitor their representatives (Arnold, 2004; Ashworth, 2012), and monitoring presupposes access: whether political news reaches a voter at all comes before what that news says. Much of the media-and-politics literature studies the second question: how voters who already receive news respond to variation in slant, framing, or coverage (DellaVigna and Kaplan, 2007; Enikolopov et al., 2011; Chong and Druckman, 2007; Chiang and Knight, 2011; Snyder and Stromberg, 2010; Myers, 2025). Studies of new-medium arrival come closer to the first: television's entry depressed turnout by displacing news consumption (Gentzkow, 2006), and early radio moved votes in Weimar Germany (Adena et al., 2015). But there the newly reached audiences already had access; Americans had radio and newspapers before television, and Germans were near-universally literate before radio. The medium changed its cost and mix, not who could consume it at all. Radio's entry into interwar America differed on exactly this margin: its audio format bypassed the literacy requirement itself, reaching voters who had never had independent access to political

information, for whom monitoring their representatives had been effectively impossible.

I examine how a new information technology reshapes electoral politics and representation when it changes the composition of the informed electorate, using radio's diffusion across interwar America (1920–1942). Radio's primary political effect, I argue, operated through reducing information poverty: the exclusion of voters from political news by literacy barriers. Its information gain was therefore largest in low-literacy communities. When these previously uninformed voters gained access, their electoral behavior changed, and their representatives' behavior changed with it.

Information access alone does not fix the direction of the shift; that depends on whose partisanship the new information unsettles. It disrupts behavior only where inherited misalignment and information deficit overlap: without misalignment there is nothing to correct, without deficit nothing to update. In the rural non-South, Republican partisanship was inherited from the Civil War era and sustained by community tradition, while the party's 1920s economic policy no longer served much of its rural base. Democratic voters fit the overlap less often, so the framework predicts a pro-Democratic disruption, not because radio favored Democrats but because the pool of misaligned, uninformed voters was concentrated in the Republican rural base.

The evidence bears this out. Using geographic determinants of AM reception — ground conductivity and woodland share, after Stromberg (2004) — as instruments for radio adoption, I find a 10-percentage-point increase in radio ownership raised non-South Democratic presidential vote share by 2.5 points, while in the South the effect collapses to near zero. Radio also depressed within-state turnout, reversing the positive association in cross-sectional work. And the vote shift carries the central clue to the mechanism: Democratic gains came from declining Republican votes, not growing Democratic ones.

Legislators responded too: where information poverty was greatest, representatives of both parties moved leftward: Democrats through adaptation and selection, Republicans through selection alone. These estimates rest on the assumption that, within state and

year, reception geography moves electoral change only through radio adoption; I probe it directly with pre-radio placebos, single-instrument estimates, and New Deal and farm-distress controls, reporting where the probes fall short.

Two further tests separate the account from its rivals. First, radio's effect on Democratic vote share scales with county illiteracy in the non-South, yet within newspaper-tracked counties it does not vary with local newspaper density (Gentzkow et al., 2011): a content-based mechanism would not generate the illiteracy gradient, and a print-substitution mechanism would not leave the effect indifferent to newspaper supply. Second, the effect is substantial in the FDR era, inviting a simpler reading: FDR's own broadcasts, not information access, moved voters; but his broadcasts reached literate and illiterate listeners alike, so that channel predicts no literacy moderation, precisely the pattern it cannot explain. The Southern null, by contrast, tests the account against its own scope condition: information poverty was if anything more severe there, but one-party Democratic dominance left too few Republican voters for an information-driven disruption to register — the deficit without the pool.

Two studies of the same medium and era anchor the comparison. Wang (2021) finds Father Coughlin's broadcasts cost Roosevelt votes in 1936, a same-era effect opposite to mine, but the two estimate different things: Wang isolates one broadcaster's content, while my design recovers the average effect of radio *access*. And the founding panel study of radio-era campaigning, Lazarsfeld et al.'s (1944) Erie County survey, found radio mostly reinforced existing preferences. The literacy gradient reconciles my larger effects with that canon: Lazarsfeld studied a literate, media-saturated county, exactly where my framework predicts the minimal effects he found.

Three contributions follow. First, a new medium's political effect can turn on its audience rather than its content: on how much of that audience the prior medium had excluded, and on the misalignment the new information exposes among them; this audience-composition contingency goes undetected in designs that track a new medium's

arrival among already-literate publics (Adena et al., 2015; Enikolopov et al., 2011) or vary the slant shown to already-informed ones (DellaVigna and Kaplan, 2007). Second, I revise the received empirical picture of radio’s effects: the turnout gains attributed to radio in cross-sectional designs (Stromberg, 2004) reflect between-state differences, not radio; within states, radio demobilized more voters than it mobilized and moved the partisan balance in the process. Third, I show how legislatures adjust to a changed electorate, not merely that they do: separating within-legislator adaptation from electoral selection, radio’s representational effects ran through both channels among Democrats but through selection alone among Republicans.

2 Theory: Information Poverty and Electoral Politics

2.1 Information Poverty and Democratic Accountability

Effective representation depends on informed citizens: voters need political information to identify their representatives, connect policy to their interests, and decide whether incumbents merit retention (Arnold, 2004; Ashworth, 2012). When they lack it, the electoral connection weakens unequally: uninformed voters can neither select well nor sanction poor performance, so legislators have less reason to answer to them. Responsiveness therefore tilts toward the informed, making information a determinant not only of whether representation works but of whose preferences it tracks.

Both halves of this relationship — information strengthens the electoral connection, and representation favors the better-informed — appear across diverse settings: U.S. legislators whose districts have fuller local newspaper coverage vote more consistently with constituents and secure more federal spending (Snyder and Stromberg, 2010; Myers, 2025), and in developing democracies newspaper circulation, press campaigns, and broadcast audits strengthen responsiveness and accountability (Besley and Burgess, 2002; Reinikka and Svensson, 2005; Ferraz and Finan, 2008). These study how variation in

the information environment translates into electoral and policy outcomes. The interwar United States poses a question one step earlier: what happens when the dominant medium itself excludes millions through the literacy barrier, and a new medium then bypasses that exclusion?

This exclusion was particularly stark. By the 1930 Census, illiteracy exceeded 10% in Southern states and 16% among African Americans nationally. These were not small or isolated populations: millions of voting-age adults could not read a newspaper, a campaign flyer, or a ballot, effectively excluded from the era's primary political medium.

Newspaper supply varied widely across the interwar United States (Gentzkow et al., 2011), but even abundant newspapers could not close the literacy gap: additional circulation meant more information for literate citizens and none for illiterate ones. This made literacy, not circulation intensity, the binding constraint on the print-information environment, the constraint radio's audio format was positioned to relax.

By "information poverty" I mean systematic exclusion from print political news due to literacy, adapting Norris (2001)'s term for unequal access to the internet's civic resources, with literacy rather than connectivity as the barrier. So defined, it is a property of *voters*, a literacy-imposed barrier to one medium, not of the local information *environment* (the abundance of news outlets); this is the distinction the newspaper-density test in Section 5 turns on. It weakens electoral accountability as described, and the framework predicts that newly informed voters will change their behavior and that representatives will change theirs in response.

2.2 Radio and the 1920s Political Order

Radio's political significance lies in reducing information poverty: its audio format removed the literacy precondition for political news. For literate voters already well-served by print it was complementary; for illiterate voters it was transformative, reaching households that had never subscribed to a newspaper and breaking a barrier that had excluded

them from political information.

Commercial broadcasting began in 1920; by the late 1920s NBC and CBS carried campaign speeches, convention coverage, and policy debates nationwide, and from 1933 Roosevelt's Fireside Chats reached tens of millions at once. By the mid-1930s roughly 60% of U.S. households owned a radio, climbing toward 90% by the early 1940s (Craig, 2000); the county mean in my sample reaches 69% by 1940. This content was predominantly national, presidential politics and policy rather than local or congressional races.

Whether the poorest of the newly reachable audience could afford a set is a fair question; early receivers were costly and adoption skewed urban and affluent. Two features close most of the gap. Listening was communal in ways print was not: neighbors, stores, and churches shared sets, so household ownership understates exposure among the poor (Craig, 2000). And the affordability constraint collapsed over the study window, as cheap AC table sets sold for under \$20 by the mid-1930s and farm penetration converged toward urban rates in precisely the FDR-era years where the effects concentrate. Within-county selection into ownership remains unobservable in a county-level design; these two features bound that concern rather than removing it.

The county-level implication is direct: radio's electoral effects should concentrate in high-illiteracy areas, where its marginal contribution to the information environment was largest.

The framework does not by itself predict the *direction* of the shift; that depends on a historical misalignment between informationally poor voters' partisan behavior and their economic interests, one that information poverty itself helped sustain.

By the 1920s the Republican Party's dominance in the rural North rested not on policy but on identity. The Civil War created deeply localized partisan cultures: communities that had sent men to fight for the Union became reliably Republican, a loyalty transmitted through family, veterans' organizations, and small-town life. Kalmoe (2020) shows county-level Union war deaths predicted Republican vote shares through at least 1912,

half a century after Appomattox. The 1896 realignment reinforced rather than replaced this inheritance: Northern farmers, more commercial and less indebted than their Southern and Plains counterparts, rejected Bryan's free silver for the Republican promise of sound money (Burnham, 1970; Sundquist, 1983). By the 1920s that rationale had decayed while the habit persisted: the party of sound money and industrial tariffs no longer answered the farm crisis.

This inherited partisanship had become a profound misalignment between partisan behavior and economic interest. For farm communities the crisis began not with the 1929 crash but a decade earlier: agricultural income collapsed by roughly two-fifths between 1919 and 1921 and never recovered during the Republican ascendancy (Saloutos, 1982). Heavily mortgaged and credit-dependent, farmers were acutely vulnerable to the tight money Coolidge and Hoover favored, and Coolidge twice vetoed the McNary-Haugen farm relief bill (1927, 1928) on grounds farm advocates read as favoring Eastern business. Republican economic policy, in its particulars, transferred wealth from agriculture to industry.

The grievance was loudly voiced through the Farm Bloc, the Nonpartisan League, and La Follette's 1924 Progressive candidacy, but largely *within* the Republican Party, whose own senators sponsored the farm relief bills (Saloutos and Hicks, 1951; Hansen, 1991). What information poverty sustained was therefore not ignorance of grievance but a failure of partisan translation: pressing discontent through one's inherited party took no informational leap, but recognizing that the *opposing* party might better serve one's interests required following platforms and legislative records that traveled almost entirely in print. That cost fell least on the literate farm belt and hardest on the non-South's low-literacy populations, where habitual Republican voting persisted by default.

Radio lowered this translation cost: for the first time a mass medium carried political debate in spoken language requiring no literacy. Interwar broadcasters were cautious about overtly partisan content outside campaign windows (Craig, 2000), so the politi-

cal payload traveled through campaign broadcasts, convention coverage, and presidential addresses. The *National Farm and Home Hour* (a USDA–NBC broadcast aired daily from 1928) carried market and farm-policy information into households print had never reached, and after 1933 explained the new programs to their intended beneficiaries (Craig, 2000). For voters whose Republican partisanship was inherited rather than chosen, radio made the gap between that inheritance and the national alternatives newly legible.

The voters most affected sat at the intersection of informational and economic precarity, and the two margins of my design map onto different parts of that population. The literacy *gradient* is identified off the non-South counties where illiteracy concentrated: the Border South and Appalachia, the Hispanic Southwest, and immigrant industrial counties. The misaligned farm belt of the rural North was, by contrast, among the most literate and print-saturated populations; for those voters radio’s marginal value ran through immediacy rather than the literacy bypass, so their response loads on the *level* effect rather than the interaction. One caveat: the Census measures literacy in any language, so in immigrant counties foreign-language-press readers count as literate, understating exclusion from the English-language print economy.

What the newly reached heard depended on when they listened — and the timing is itself a prediction. Through most of the 1920s no aligned alternative existed: the Democratic Party of John W. Davis and Al Smith offered Eastern, urban candidacies with no programmatic answer to farm distress, so radio had little misalignment-revealing content to carry before 1932. The aligned alternative arrived with Roosevelt’s 1932 campaign and the New Deal: farm price supports, labor protections, banking regulation, and relief spending (Leuchtenburg, 1963; Cohen, 1990). Radio’s role was to make it audible to voters print could not reach. The mechanism therefore requires information *and* an aligned alternative; since the alternative arrived in 1932, the framework predicts that radio’s pro-Democratic effect should *concentrate* in the FDR era rather than appear uniformly across the 1920s, a prediction the era split in Table 7 bears out.

The same account identifies where radio's effect should *not* appear. For observable electoral change there must be a population whose voting behavior can shift in response to new information, a scope condition the interwar South fails on two distinct margins.

First, among white voters the partisan structure left no population susceptible to disruption. The few white Republicans clustered in Appalachian mountain counties such as East Tennessee, Western North Carolina, and West Virginia, where Unionist Civil War heritage sustained a minority tradition; even if radio informed them, the pool was too small to move aggregates. The white Democratic majority was bound by regional and racial identity forged in the Civil War and Reconstruction: Democratic presidential vote share already averaged 0.77 across Southern counties, and where economic policy entered, especially under the New Deal, it reinforced that allegiance. For these voters radio revealed no misalignment.

Second, for Black Southerners, a large share of the region's informationally poor, Jim Crow disenfranchisement severed the link between information and voting: poll taxes, discriminatory literacy tests, white-only primaries, and the threat of violence and economic retaliation (Key, 1949; Katznelson, 2013) ensured that even informed listeners could not act at the ballot box.

Together, these conditions generate a graduated rather than binary prediction: radio's effect should scale with the size of the local misaligned pool. Counties with substantial Republican minorities should show the largest Democratic gains; those with near-zero Republican vote shares, near-zero effects regardless of information poverty. The Southern null should therefore appear as a floor effect across all outcomes, not a selective failure of the information channel.

2.3 Behavioral Predictions and Competing Mechanisms

When previously uninformed voters encounter political information for the first time, their responses need not be uniform. I distinguish three types of behavioral change that

an information shock can produce, each with distinct aggregate implications.

Switching: existing voters change allegiance, learning that the opposing party's platform better aligns with their economic interests than the party they had supported out of habit, community norm, or inherited identity. *Demobilization*: existing voters withdraw, when new information creates dissonance with their partisan attachment but the costs of switching are too high. *Mobilization*: previous non-voters, now informed enough to form a preference, enter the electorate.

These three responses, mutually exclusive and collectively exhaustive of an individual voter's behavioral margins, generate distinct aggregate predictions depending on how they combine. In the interwar context the shock was asymmetric: radio made visible the misalignment between Republican economic policy and low-literacy voters' interests while largely confirming Democratic voters' existing alignment. Switching and demobilization should therefore fall disproportionately on Republicans; mobilization toward Democrats should occur but may be partly offset by demobilization of right-leaning habitual Democrats under similar pressure. The aggregate prediction is correspondingly asymmetric: an unambiguous decline in Republican counts, an ambiguous net change in Democratic counts, and a net pro-Democratic shift in vote shares, because the misaligned were concentrated among habitual Republicans. Turnout, balancing mobilization against demobilization, is likewise ambiguous. The framework thus predicts sharply for vote share but ambiguously for turnout; I formalize the decomposition in the Appendix.

These predictions follow from a single channel, information-poverty reduction; but several alternatives could produce similar aggregate patterns, and the question is whether the information channel dominates. *Salience mobilization* holds that radio drew disengaged citizens to the polls by making politics entertaining and accessible, regardless of policy learning. *Rhetorical persuasion* emphasizes spoken content's power over voters who already had print information, distinct from persuasion of newly *reached* listeners, taken up below. *Entertainment crowd-out* holds that radio privatized leisure and displaced

Table 1: Competing Mechanisms and Testable Predictions

	Literacy interaction	Newspaper-density independence	FDR-era uniqueness	Legislator response	Asymmetric Rep vote loss
Information poverty reduction	Yes	Yes	No	Bipartisan	Yes
Salience mobilization	Yes (symmetric)	No	No	Unclear	No
Persuasion (of print-informed)	No	No	Depends	Bipartisan	Weak
Entertainment crowd-out	Turnout only	No	No	None	No
FDR-specific messaging	No	No	Yes	Bipartisan or asymmetric	Unclear
Elite signaling	n/a	n/a	Yes	Asymmetric	No
<i>Observed (this paper)</i>	Yes (1.62*)	Yes (≈ 0)	Ambiguous	Bipartisan	Yes (Rep -1.07^{***})

Note: Rows are candidate mechanisms; columns are testable dimensions, each evaluated in the indicated table (literacy interaction, Table 6; newspaper-density independence, Table 8; FDR-era uniqueness, Table 7; legislator response, Table 5; asymmetric Republican vote loss, Table 4). Entertainment crowd-out predicts a literacy gradient on turnout only; salience mobilization predicts symmetric positive moderation by both literacy and newspaper density; elite signaling operates on legislators rather than voters. The newspaper-density cell (≈ 0) refers to per-capita circulation, while count and total-circulation measures carry small positive interactions about one-eighth the literacy gradient (Appendix Table A20).

civic engagement, depressing turnout without transmitting information at all (Gentzkow, 2006; Olken, 2009; Kim and Patterson, 2025). *FDR-specific messaging* attributes the pro-Democratic effect to Roosevelt’s content rather than a general information mechanism. *Elite signaling* posits that legislators responded to party-leadership signals via radio rather than to voters, predicting a *party-asymmetric* response. Each generates distinct predictions; Table 1 summarizes them.

The literacy gradient is the leading discriminator but cannot stand alone: both information poverty and salience mobilization predict larger effects where illiteracy is higher. They diverge on newspaper circulation: literacy was an absolute barrier, circulation only a difference of degree among those who could already read. So information poverty predicts radio’s effect tracks illiteracy far more strongly than newspaper density, while salience mobilization should also respond to thin newspaper supply. This *asymmetry*, a steep illiteracy gradient alongside a weak newspaper-density one, is the separating test, not the literacy gradient alone. The remaining rivals make distinct predictions (no literacy gradient for persuasion or FDR-specific messaging; a turnout-only effect for crowd-out; a party-asymmetric legislator response for elite signaling), taken up in turn.

3 Research Design and Data

3.1 Identification Strategy

A naive regression of political outcomes on radio ownership would confound radio's effect with the characteristics of communities that adopted it early: wealthier, more urban, more politically engaged places may have adopted radio faster while experiencing different political trends for unrelated reasons. I address this selection bias with instrumental variables (IV) that exploit geographic determinants of radio reception quality.

Building on Stromberg's (2004) cross-sectional instruments, I interact two time-invariant geographic features, ground conductivity and woodland share, with time, generating within-county variation in predicted radio adoption. The logic is simple: in the early 1920s, with few stations broadcasting, reception quality mattered little; by the late 1920s and 1930s, when national networks offered hours of daily programming including political content, the same geographic differences translated into meaningful variation in ownership and listening. The time interaction captures these increasing returns to reception quality as the industry matured, within-county variation a purely cross-sectional design would lack.

My primary specification is county-level two-stage least squares with county and state-by-year fixed effects. For county c in state s and election year t :

First stage:

$$\text{Radio}_{ct} = \alpha_c + \gamma_{st} + \beta_1(\text{GroundConductivity}_c \times t) + \beta_2(\text{WoodlandShare}_c \times t) + \mathbf{X}'_{ct} \boldsymbol{\delta} + \epsilon_{ct} \quad (1)$$

Second stage:

$$Y_{ct} = \alpha_c + \gamma_{st} + \theta \cdot \widehat{\text{Radio}}_{ct} + \mathbf{X}'_{ct} \boldsymbol{\lambda} + u_{ct} \quad (2)$$

where Y_{ct} is the electoral outcome: presidential Democratic two-party vote share, presidential turnout, House Democratic two-party vote share, or log total House votes.

County fixed effects α_c absorb all time-invariant county characteristics (geography, settlement patterns, persistent political culture, long-run economic structure); state-by-year fixed effects γ_{st} absorb all state-level time-varying confounders such as state economic trends, gubernatorial elections, and policy changes, on top of national trends.

The control vector $\mathbf{X}_{ct}$ includes urbanization, Black population share, and log county population, addressing the concern that radio adoption correlates with demographic and economic changes that independently affected voting. Standard errors cluster at the county level (district level for the district analysis), matching where radio exposure varies; for interactions with county illiteracy, whose identifying variation is largely between-state, I report the more conservative state-clustered errors. Because the instruments are spatially smooth fields interacted with time, county clustering may also understate level-specification errors, so Appendix Table A9 reports every headline level estimate under both clusterings (the non-South share, turnout, and Republican-count results survive state clustering; the supplementary full-sample share estimates do not). Weak-instrument and overidentification diagnostics — partial R^2 , Kleibergen–Paap statistics, Hansen J tests, Sanderson–Windmeijer conditional F — appear in Appendix Table A3 and the table notes.

The key identifying assumption is the exclusion restriction: conditional on county and state-by-year fixed effects and controls, the interaction of geographic radio propagation features with time affects electoral outcomes only through its effect on radio adoption. This is a non-trivial assumption that warrants careful discussion. County fixed effects absorb the direct influence of geography on political outcomes.

The assumption would be violated if geographic reception advantages translated into political change over time through channels other than radio. The two instruments differ here. Ground conductivity reflects subsurface geology, integrated far below the topsoil that governs agricultural productivity, so the exclusion restriction is most plausible for it. Woodland share is a more direct concern: heavily forested counties had different eco-

economic structures, lower densities, and potentially different trajectories independent of radio. The design cannot lean on the cleaner instrument alone, since the two identify jointly, and within the primary state-by-year design woodland carries the larger partial R^2 (0.15 versus 0.04 in the non-South). I therefore assess the concern directly with pre-radio placebos, conductivity-only estimates, and Hansen J tests of joint validity (Appendix Tables A4, A6, A3), rather than by instrument weighting.

State-by-year fixed effects provide a further safeguard, identifying the effect from within-state, within-year variation alone. Because my instruments carry both within- and between-state variation, this removes the between-state component, absorbing much of the instrument variance, yet first-stage F-statistics remain above 600 in the main samples, confirming within-state variation alone identifies strongly. I treat it as primary because between-state exposure correlates with state economic development, urbanization, and competitiveness, all of which independently shaped electoral trends; county-and-year estimates are reported alongside.

3.2 Data Sources

County-level radio ownership comes from the 1930 and 1940 U.S. Censuses (the first to ask about radio sets), with ground conductivity and woodland share from Stromberg (2004); ownership in 1920 is set to zero, as commercial broadcasting began only in November 1920. Measured as the fraction of households owning a receiver, diffusion was rapid: across my 3,000-plus counties, mean ownership rose from zero in 1920 to 26% in 1930 and 69% in 1940, surpassing daily newspapers as the most widely used political medium by the mid-1930s. I linearly interpolate to non-Census election years, so identification rests on the cross-decade Census variation; because the interpolation error is a systematic function of each county's adoption path rather than classical noise, I do not use it to adjudicate IV-versus-OLS magnitudes.

County-level presidential returns come from Clubb et al. (2006) (Democratic two-party

vote share and turnout, six elections 1920–1940); House returns from the United States Historical Election Returns collection (Inter-university Consortium for Political and Social Research, 2025) (Democratic two-party share and total votes, eleven elections 1920–1942). I compute two-party share as the Democratic share of the combined Democratic and Republican vote, excluding third parties for comparability across elections, avoiding conflating two-party competition with third-party movements such as La Follette’s 1924 Progressive candidacy.

Turnout is total votes cast divided by estimated voting-age population per county-year. Because county population is decennial, I interpolate between Censuses, introducing denominator measurement error unlikely to correlate with radio penetration conditional on my controls.

County illiteracy, the Census measure for persons aged 10 and over unable to read or write in any language, comes from the 1920 and 1930 Censuses; persons literate only in a non-English language count as literate, so it understates exclusion from the English-language press. The distribution was sharply uneven: mean county illiteracy was 5.3% in 1920 and 4.1% in 1930, but Southern states exceeded 10% versus below 4% in the North, with substantial within-state variation (in Texas, near zero to 44%). I assign it as a two-step function of election year, breaking at 1925 (the 1920 rate for 1920–1924 elections, the 1930 rate thereafter).¹

County-level daily newspaper circulation comes from the United States Newspaper Panel (Gentzkow et al., 2011), tracking individual local papers’ daily circulation across counties and years. I construct three county-year measures: distinct-paper count, total daily circulation, and circulation per capita. I use circulation per capita, the access-intensity measure, as the primary moderator in Section 5 (Appendix Table A20 reports the others). Of roughly 3,100 sample counties, 1,168 (35%) had a tracked daily at some

¹Unlike radio ownership (linearly interpolated), illiteracy uses a step function: the 1940 Census replaced the read/write question with “highest grade of school completed,” leaving no terminal value to interpolate toward, so I carry the 1930 rate forward to the 1940 election. The cross-Census county correlation is 0.94, so the geographic pattern of illiteracy was highly stable across decades.

point during 1920–1940; the rest had none, so county fixed effects leave nothing to identify there. The newspaper analysis therefore uses the ever-tracked counties (1,067 enter after merging and dropping singletons; Ns in Table 8). Summary statistics: Appendix Table A1.

Legislator ideology scores come from the Voteview project (Lewis et al., 2023). I use two complementary measures on the first (liberal-conservative) dimension. The Nokken-Poole score is re-estimated each Congress and so varies within a legislator’s career, capturing both adaptation (within-legislator change in roll-call behavior) and selection (replacement of incumbents). The common-space DW-NOMINATE score fixes each legislator’s position across their career, so within-district changes reflect only selection. Comparing the two separates the adaptation and selection channels; I also add legislator fixed effects to the Nokken-Poole score to isolate the within-legislator adaptation margin on a single scale (Appendix Table A23).

For the district-level analysis, I aggregate county radio data to congressional districts using population-weighted area crosswalks (Ferrara et al., 2024), which handle the complex county-to-district mapping that shifts with each decennial redistricting, weighting each county by its share of district population to give a population-representative measure of district radio exposure.

The instruments and key variables are sharply patterned geographically: ground conductivity is highest in the Great Plains and woodland share in the Southeast and Pacific Northwest, both with substantial within-state variation, the identifying variation that survives state-by-year fixed effects, while 1930 radio ownership mirrors conductivity and illiteracy concentrates in the South and Appalachia (Figure 1).

3.3 Sample

The merged dataset is a county-year panel in which nearly all counties appear in every election year, with some entry and exit from boundary changes, missing returns, or absent

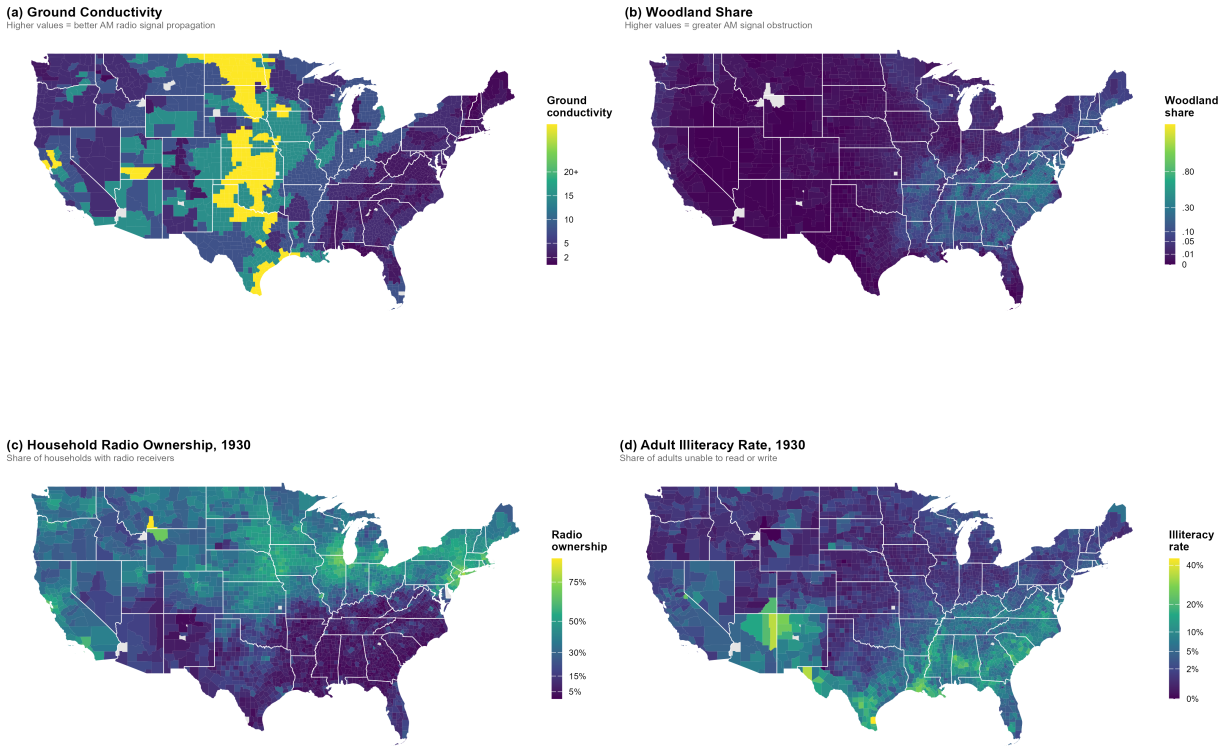


Figure 1: Geographic Variation in Instruments and Key Variables

Note: County-level maps of the two instrumental variables (ground conductivity and woodland share), radio ownership (1930 Census), and illiteracy rate (1930 Census). Ground conductivity measures the quality of AM radio signal propagation; higher values indicate better reception. Woodland share measures signal obstruction from tree cover. The county-level variation visible within states is the identifying variation that survives state-by-year fixed effects.

Census data; county boundaries were largely stable across 1920–1940.

The presidential panel includes approximately 18,300 county-year observations across more than 3,000 counties and six elections (1920–1940); the House panel is larger, approximately 34,300 county-year observations across eleven elections (1920–1942), reflecting the biennial cycle.

Conventional first-stage F-statistics far exceed the Stock and Yogo (2005) weak-instrument critical values under both fixed-effect structures, and cluster-robust Kleibergen–Paap statistics remain far above conventional thresholds in the main samples (Appendix Table A3). The district-level analysis uses 4,304 legislator–Congress observations from

Table 2: Radio and Democratic Vote Share

	(1) Full	(2) Non-South	(3) South
<i>Panel A: Presidential elections</i>			
Radio	0.092 ⁺ (0.051)	0.250*** (0.040)	−0.109 (0.132)
First-stage F	685	1,000	109
N	18,317	11,689	6,628
<i>Panel B: House elections</i>			
Radio	0.103 ⁺ (0.061)	0.181** (0.056)	−0.020 (0.121)
First-stage F	1,248	1,711	249
N	34,314	21,716	12,598
County FE	Yes	Yes	Yes
State × Year FE	Yes	Yes	Yes

Note: IV estimates with county and state-by-year fixed effects. Dependent variable is Democratic two-party vote share. Controls include urbanization, Black population share, and log population. Instruments are ground conductivity × time and woodland share × time. County-and-year fixed effects estimates are reported in Appendix Table A5. Significance based on county-clustered SEs: ⁺ $p < 0.10$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

the 67th through 76th Congresses (1921–1941).

4 Results

4.1 Main Electoral Effects

Presidential Vote Share I begin with presidential elections, where radio’s informational effect should be strongest given the national focus of interwar programming. The framework predicts that radio should have raised Democratic vote share in regions where the misaligned-uninformed pool was concentrated (primarily the rural non-South), with effects strongest where information poverty was most severe. Table 2 presents IV estimates across regional subsamples.

Consistent with the theory, radio increased Democratic presidential vote share. Under state-by-year fixed effects, which restrict identification to within-state reception vari-

ation, the non-South effect is substantial and highly significant (Table 2, column 2): a 10-percentage-point increase in radio ownership corresponds to roughly 2.5 points of Democratic vote share. The county-and-year specification yields a much larger estimate (Appendix Table A5), but that gap reflects between-state Democratic vote growth more plausibly driven by state-level trends correlated with radio than by radio itself. Adding state-specific linear time trends to the county-and-year specification, a far milder control, reproduces the within-state estimate almost exactly (Appendix Table A7). I therefore treat within-state estimates as primary.

The magnitude is large but consistent with the media-and-voting literature, where extending access to audiences that lacked an accessible source moves votes far more than shifting the slant of a source they already consume (Enikolopov et al., 2011; DellaVigna and Kaplan, 2007; Adena et al., 2015; Martin and Yurukoglu, 2017).

The regional pattern confirms the scope condition from the Theory section. In non-Southern counties², where a substantial Republican pool existed, the effect is more than double the full-sample estimate. The South is the mirror image: the effect collapses to statistical zero even though the first stage remains strong, because Southern Democratic vote share already averaged 0.77, leaving a negligible Republican pool, while Jim Crow disenfranchisement kept mobilization from the ballot box.

House Elections The House results (Panel B of Table 2) show a similar non-South pattern, significant and roughly three-fourths the presidential magnitude: smaller, as expected, given the presidential focus of interwar network programming; individual House races remained the province of local newspapers. The Southern House estimate is again indistinguishable from zero, mirroring the presidential result.

²I define the South as the eleven former Confederate states: Alabama, Arkansas, Florida, Georgia, Louisiana, Mississippi, North Carolina, South Carolina, Tennessee, Texas, and Virginia.

Table 3: Radio and Voter Participation

	(1) Full	(2) Non-South	(3) South
<i>Panel A: Presidential turnout (fraction of eligible population)</i>			
Radio	−0.125** (0.042)	−0.322*** (0.037)	0.192+ (0.109)
First-stage F	686	999	110
N	18,281	11,673	6,608
<i>Panel B: House turnout (fraction of county population)</i>			
Radio	−0.032 (0.029)	−0.173*** (0.034)	0.138* (0.060)
First-stage F	1,252	1,638	249
N	34,904	22,321	12,583
County FE	Yes	Yes	Yes
State × Year FE	Yes	Yes	Yes

Note: IV estimates; controls, instruments, and significance conventions as in Table 2. Panel A dependent variable: total presidential votes divided by ICPSR voting-age population, capped at 1.0 (binds in 39 county-years with undercounted denominators); 6 election years. Panel B: total House votes divided by county population (no House-specific voting-age estimate exists); 11 election years, odd years excluded. County-and-year fixed-effects estimates in Appendix Table A5.

Turnout and Vote Count Decomposition Table 3 presents turnout results. Under state-by-year fixed effects, radio *decreases* presidential turnout in the non-South, highly significantly.³ In the South the point estimate is marginally positive: in tension with, but not significantly contradicting, the scope-condition prediction of a clean Southern null.

To understand the negative turnout effect, I decompose total votes into Democratic and Republican components. Table 4 reports non-South IV estimates for log Democratic, log Republican, and log total votes under state-by-year fixed effects.

The decomposition reveals a striking asymmetry at the non-South average (Table 4). Within states, radio did not increase Democratic vote counts, leaving them statistically indistinguishable from their pre-radio level, but sharply and significantly reduced Republican counts. The Democratic vote-share increase therefore came not from new Democratic

³Turnout (votes cast over voting-age population, VAP) exceeds an impossible 100% in 39 county-years (0.21% of obs) where ICPSR's VAP denominator is undercounted, mostly independent cities whose votes are coded to the city while VAP comes from another unit. I cap these at 1.0; without the cap the non-South coefficient remains negative and highly significant, the cap reducing its magnitude by roughly 18%.

Table 4: Vote Count Decomposition (Non-South, State $\times$ Year FE)

	Log Dem	Log Rep	Log Total	Dem Share	Turnout
Radio	0.039 (0.170)	-1.069*** (0.116)	-0.438*** (0.071)	0.250*** (0.040)	-0.322*** (0.037)
County FE	Yes	Yes	Yes	Yes	Yes
State $\times$ Year FE	Yes	Yes	Yes	Yes	Yes

Note: IV estimates, non-South sample. Controls, instruments, and significance conventions as in Table 2. Standard errors clustered at county level in parentheses. * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

voters but from Republican voters withdrawing, and the negative turnout effect follows mechanically. This asymmetry is consistent with the mechanism: habitual Republican voters, whose behavior had persisted through information isolation, were disrupted by radio’s shock. The near-zero change in Democratic votes does not imply no Democratic-side flows; as the formal decomposition shows (Appendix D.1), it can reflect cancellation between inflows (mobilization, switching from former Republicans) and outflows (departure of right-leaning habitual Democrats). In the South, neither party’s votes moved significantly (Appendix Table A13), consistent with the scope condition.⁴

The decomposition also speaks to a long-standing question about the New Deal realignment: whether the Democratic majority was built by converting Republicans or mobilizing new voters (Andersen, 1979; Erikson and Tedin, 1981). My within-state estimates isolate a third channel neither pole anticipated: a Democratic-share gain with *no* growth in Democratic votes, produced by rural Republican demobilization, which encompasses rather than complements the urban-mobilization thesis (Appendix D.3).

A caveat applies. Because my data are aggregate county returns, “switching” and “demobilization” describe aggregate patterns rather than observed individual transitions; the decomposition establishes what happened to each party’s vote totals, not the

⁴The county-and-year positive turnout finding (replicating Stromberg, 2004) reflects between-state Democratic vote growth that disappears within state, alongside a persistent Republican decline (Appendix Table A12). The turnout placebo supports treating state-by-year as primary: ground conductivity $\times$ time predicts pre-radio turnout (1908–1916) under county-and-year but not state-by-year; the Democratic-share placebo is at most marginal and negative-signed, against the headline (Robustness section; Appendix Table A4).

individual-level behavioral mechanism.

With that caveat, the competition split (Appendix Table A14) illuminates the relative contributions. Where Republicans held safe 1920 majorities, radio produced a significant Democratic vote increase alongside the Republican decline, consistent with switching. In competitive counties, Democratic votes did not increase while Republican votes still declined, consistent with demobilization. And where Democrats held safe 1920 majorities (a thin cell of 66 counties), the misaligned pool was composed of habitual Democrats rather than Republicans, and the Democratic-share effect vanishes, the mirror image of the safe-Republican pattern. The vote share effect declines monotonically across the three competition bins, matching the scope condition: radio's effect is larger where the misaligned pool is larger.

The competition split doubles as the direct test of entertainment crowd-out (Table 1): crowd-out predicts depressed participation wherever radio diffused with no directional vote movement, so it cannot generate the significant Democratic vote *increase* in safe-Republican counties, and it predicts a Southern turnout decline where the point estimate is instead positive (Table 3).

Two caveats bound these magnitudes. All estimates are within-state relative effects: comparisons across counties with different reception geography in the same state and year, not aggregate time-series predictions, so a negative relative effect is fully compatible with the era's rising national turnout, which the fixed effects absorb. And the IV identifies a local average effect for counties whose adoption reception geography shifted; extrapolating to the full 1920–1940 diffusion path would overstate what the design identifies.

4.2 Legislator Ideology

The behavior of elected representatives is the final link in the accountability chain, and the least guaranteed to follow from a shift in votes: incumbents can outlast a changing

electorate, and replacement and adaptation may be gradual and noisy. Whether radio's information shock propagated this far is a distinct question, taken up at the district level: did a disruption that informed previously excluded voters also register in the representatives those districts sent to Congress?

Using population-weighted crosswalks (Ferrara et al., 2024), I aggregate county radio data to congressional districts and estimate IV regressions with district and Congress fixed effects, using the same geographic instruments as my county-level specification.⁵

Table 5 presents the two complementary ideology measures: Panel A the Nokken-Poole first-dimension score, re-estimated each Congress and so capturing adaptation and selection; Panel B the DW-NOMINATE first-dimension score, fixed over each legislator's career and so capturing selection only. Negative values indicate more liberal positions.

Radio produced a substantial leftward shift in district-level legislator ideology (Table 5): the full-sample estimate is large, precise, and highly significant under both measures. That the Nokken-Poole estimate (Panel A) exceeds the DW-NOMINATE one (Panel B) in absolute value is a first hint that adaptation operates on top of selection. One caveat throughout: because state-by-Congress fixed effects are infeasible at the district level, these estimates rest largely on between-state instrument variation of the kind the county analysis treats as potentially confounded, so I read Table 5 primarily for its sign pattern and bipartisan structure, not its magnitudes.⁶

Both Democratic and Republican legislators in high-illiteracy districts shifted leftward. In Panel B, selection only, they shifted by similar magnitudes in the same direction, both significantly. This bipartisan pattern is the prediction the framework entered into Table 1: because the Democratic program better served the misaligned voters radio newly

⁵State-by-Congress fixed effects are infeasible: state-by-year fixed effects absorb 79–84% of district-level instrument variation, collapsing the first stage (county-level absorption shares in Table A2). I split the sample by district illiteracy via separate subsample regressions; a fully saturated interaction IV, equivalent to the split, is used only to compute the high-minus-low difference.

⁶Pushing further, census-division-by-Congress fixed effects, absorbing 56–57% of residual instrument variance beyond Congress fixed effects, preserve negative full-sample point estimates but render them imprecise, while the illiteracy and party subsample first stages weaken enough to be uninformative rather than contradictory (Appendix Table A22).

Table 5: Radio and Legislator Ideology (District Level)

	(1) Full	(2) High Illit	(3) Low Illit	(4) Dem, Hi Illit	(5) Rep, Hi Illit
<i>Panel A: Nokken-Poole first dimension (adaptation + selection)</i>					
Radio	-1.794*** (0.234)	-1.550*** (0.281)	-1.203 (0.944)	-1.033*** (0.263)	-0.286 (0.256)
First-stage F	530	307	101	123	70
N	4,304	2,149	2,149	1,565	545
Hi – Lo difference (saturated interaction IV): -0.347 (0.983), $p = 0.72$					
<i>Panel B: DW-NOMINATE first dimension (selection only)</i>					
Radio	-1.536*** (0.208)	-1.412*** (0.247)	-0.320 (0.846)	-0.726*** (0.180)	-0.669** (0.244)
First-stage F	528	306	101	123	69
N	4,290	2,141	2,143	1,560	543
Hi – Lo difference (saturated interaction IV): -1.092 (0.881), $p = 0.22$					
District FE	Yes	Yes	Yes	Yes	Yes
Congress FE	Yes	Yes	Yes	Yes	Yes

Note: IV estimates with district and Congress fixed effects; district-clustered standard errors in parentheses. Panel A uses the Nokken-Poole score; Panel B the DW-NOMINATE first-dimension score. Negative values indicate more liberal positions. Columns (4)–(5) restrict to Democratic and Republican legislators within high-illiteracy districts. The “Hi – Lo difference” row reports the radio $\times$ high-illiteracy coefficient from a fully saturated interaction IV, by construction equal to the column (2)–(3) difference. * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

informed, the shock moved the local electorate leftward, giving *both* parties’ representatives an incentive to track it rather than reflecting any party-specific message. Single-party elite signaling, by contrast, predicts asymmetric responses, since signals from one party’s leadership should not move opposing-party legislators in parallel.

The two parties responded through different channels. A within-legislator specification that absorbs incumbent replacement identifies adaptation alone: among high-illiteracy Democrats it is negative and significant ($-\$0.61$, $p < 0.05$). Sitting incumbents moved their own roll-call behavior leftward, beyond being replaced by more liberal successors. Among high-illiteracy Republicans it is indistinguishable from zero, so their response operated through selection alone (a pooled radio-by-party interaction confirms the difference, $p = 0.048$; Appendix F).

That asymmetry also closes a *two-party* signaling account: it would have to posit that

Republican elites signaled future challengers but not sitting incumbents, an unmotivated asymmetry, whereas the electorate account derives the split from brand cost: following a leftward electorate was low-cost for Democrats but meant adopting opponents' positions for Republicans (Appendix F).

Beyond corroborating the county-level results, the legislator evidence is a finding in its own right. The shock did not stop at the ballot box: it moved representatives of both parties and, among Democrats, the roll-call behavior of sitting incumbents, not only who held the seat. That a changed electorate reshapes not just who is elected but how incumbents vote is the hardest link in the accountability chain to demonstrate. It extends Snyder and Stromberg's (2010) finding that fuller local press coverage makes legislators more responsive: radio carried that accountability to voters print had never reached. And a bipartisan shift running through party-specific channels is far harder to attribute to an omitted variable than the vote-share shift alone.

5 Mechanism: The Literacy Channel

The main results are consistent with information poverty, but several alternatives in Table 1 could produce similar aggregate effects. The decisive test is whether radio's effect scales with the severity of information exclusion: if the mechanism is information poverty, radio should matter most where voters were most cut off from print.

I test this three ways: the radio-by-illiteracy interaction, temporal heterogeneity across the pre-FDR and FDR periods, and a radio-by-newspaper-circulation interaction. My primary estimator is IV, instrumenting every radio term, interactions included, with the corresponding geographic-instrument interactions.⁷ Because the scope condition restricts

⁷Using OLS for the mechanism while instrumenting the level would let a confound enter through the interaction. Interaction first-stage F-statistics are well above weak-instrument thresholds throughout (Table 6 note; Table 8, exceeding 740 per interaction in the non-South and 800 in the full sample). For the temporal test, era-by-era IV is infeasible because the FDR-era first stage weakens substantially ($F = 11.7$ full, $F = 37$ non-South) once radio saturates in the 1930s; I instead estimate a single joint-interaction IV treating radio and radio $\times$ FDR era as endogenous, using four instruments. The interaction also faces a more demand-

Table 6: Radio $\times$ County Illiteracy Interaction: IV and OLS

	Non-South (scope-relevant)			Full sample (supplementary)		
	Pres Dem Share	Pres Turnout	House Dem Share	Pres Dem Share	Pres Turnout	House Dem Share
<i>Panel A: IV interaction (primary)</i>						
Radio	0.291** (0.107)	-0.327*** (0.060)	0.222* (0.091)	0.095 (0.079)	-0.102 (0.099)	0.125 (0.079)
Radio $\times$ Illiteracy	1.620* (0.635)	-1.421** (0.542)	2.176*** (0.585)	0.208 (0.929)	-0.142 (0.522)	0.965 (0.679)
<i>Panel B: OLS interaction (supplementary)</i>						
Radio	0.162*** (0.039)	-0.079+ (0.041)	0.150** (0.046)	0.177*** (0.038)	-0.043 (0.030)	0.103*** (0.029)
Radio $\times$ Illiteracy	1.074** (0.396)	-0.261 (0.290)	1.835*** (0.490)	0.634* (0.261)	0.002 (0.166)	0.946* (0.368)
County FE	Yes	Yes	Yes	Yes	Yes	Yes
State $\times$ Year FE	Yes	Yes	Yes	Yes	Yes	Yes
N	11,689	11,673	21,716	18,317	18,281	34,314

Note: Panel A: IV with two endogenous regressors (radio, radio $\times$ illiteracy) and four instruments (ground conductivity $\times$ time, woodland $\times$ time, each also interacted with county illiteracy). Non-South first-stage F: 717 (radio), 1,737 (interaction); Sanderson-Windmeijer conditional F: 162 and 73 (county clustering), 106 and 46 (state). The weak-instrument-robust subset Anderson-Rubin 95% confidence set for the non-South interaction is [0.02, 3.58], and the Anderson-Rubin joint test rejects at $p = 0.003$ (state-clustered). Panel B: OLS analogue. Both panels include county illiteracy at level. County and state-by-year fixed effects; illiteracy time-varying by nearest Census (1920 rates through 1924, 1930 thereafter). Non-South = Confederate-11 (see Table 2 note). State-clustered SEs. + $p < 0.10$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

the mechanism to the non-South, I treat the non-South IV interaction as primary, with full-sample IV and OLS supplementary.

County Illiteracy Interaction The first test instruments radio and radio $\times$ illiteracy with the two geographic instruments and their illiteracy interactions.

The non-South IV interaction matches the framework's central prediction across all three primary outcomes: radio's effect on presidential Democratic vote share scales positively and significantly with county illiteracy; the House interaction is larger and more precise, extending the gradient beyond presidential contests; and the turnout interaction is negative and significant, refining the earlier turnout result: radio's turnout depression concentrates in high-illiteracy counties. The OLS interactions (Panel B) are same-signed

ing identification requirement than the level: standard radio-adoption confounders (income, urbanization, industrial composition) operate at the level, not differentially by the moderator.

but smaller, mirroring the OLS-below-IV ordering at the level. The full-sample interactions are same-signed but individually insignificant, the expected power cost of pooling Southern counties where the scope condition fails. Radio's effect on presidential Democratic vote share is about a quarter larger in counties at the 90th percentile of illiteracy (5.1 percent) than in fully literate ones, and remains positive and significant across the observed range.

This gradient is plausible because county illiteracy proxies a broader community gradient: for every adult who could not read at all, several more read haltingly and rarely followed newspaper policy coverage, so the interaction reflects community-level information poverty, not merely a direct effect on the completely illiterate.

Decomposing the IV illiteracy interaction by outcome (Appendix Table A16), the gradient on log Democratic votes is too noisy to identify under IV, while the gradient on log Republican votes is negative and significant: the share gradient is driven primarily by differential Republican decline at high illiteracy. This matches the formal decomposition (Appendix D.1), where the near-zero change in average Democratic counts reflects cancellation between inflows and outflows, leaving the Republican-side gradient as the cleanest count-level evidence.

A median-split alternative yields an apparent reversal: the share effect is larger in the low-illiteracy half. But this reflects sample composition (border-state counties, especially Kentucky, dominate the high-illiteracy cells), not a contradicted gradient; I treat the continuous interaction as primary and document the diagnosis in Appendix E.1.

This literacy-graded pattern is inconsistent with persuasion of print-informed audiences and with strict FDR-specific messaging (Table 1), neither of which predicts a literacy-conditional response. It does not by itself adjudicate the rest: salience mobilization also predicts positive literacy moderation (separated by the newspaper-density asymmetry below); the persuasion-of-newly-reached complement (Zaller, 1992) shares the gradient (distinguished by the demobilization-dominant count evidence); elite sig-

Table 7: Radio and Presidential Outcomes by Era

	Non-South IV (primary)		Full-sample IV (supp.)		Non-South OLS (supp.)	
	Pres Dem Share	Pres Turnout	Pres Dem Share	Pres Turnout	Pres Dem Share	Pres Turnout
Radio (pre-FDR base)	−0.134 (0.186)	−0.178 (0.108)	−0.221 (0.190)	0.035 (0.121)	0.017 (0.053)	0.079* (0.039)
Radio × FDR era	0.308** (0.102)	−0.115* (0.058)	0.249* (0.113)	−0.127* (0.053)	0.131** (0.044)	−0.177*** (0.033)
Implied FDR-era effect	0.174	−0.292	0.029	−0.092	0.148	−0.098
First-stage F (radio)	526	525	358	358	—	—
First-stage F (radio × FDR)	674	672	486	487	—	—
County FE	Yes	Yes	Yes	Yes	Yes	Yes
State × Year FE	Yes	Yes	Yes	Yes	Yes	Yes
N	11,689	11,673	18,317	18,281	11,689	11,673

Note: IV columns estimate a single regression with radio and radio × FDR-era endogenous and four instruments (ground conductivity × time, woodland × time, each also interacted with the FDR-era indicator); per-column conventional first-stage F-statistics are reported. The non-South is the scope-relevant primary sample; the full sample is supplementary. Pre-FDR = elections of 1920, 1924, 1928; FDR era = 1932, 1936, 1940 (the November 1932 campaign already referenced the New Deal program). Era-specific subsample IVs are not reported because the FDR-era first stage weakens substantially ($F = 11.7$ full; $F = 37$ non-South); the joint-interaction era estimates need not average to the pooled Table 2 estimates (different instrument set, different complier weights). OLS columns report the analogous specification without instrumenting, on the IV sample. $^+p < 0.10$, $^*p < 0.05$, $^{**}p < 0.01$, $^{***}p < 0.001$.

naling is addressed by the bipartisan legislator response (Section 4.2); and a milder, Roosevelt-era-only version of FDR-messaging survives until the temporal evidence below.

Temporal Heterogeneity A natural concern is that radio’s pro-Democratic effect reflects the New Deal context rather than a general information mechanism. The joint-interaction IV above recovers pre-FDR and FDR-era effects from a single specification, with the analogous OLS era interaction for comparison.

Table 7 presents the joint-interaction IV era split (non-South primary; full-sample IV and OLS supplementary). Under the non-South IV, the radio × FDR-era coefficient on Democratic vote share is positive and significant at the 1% level and the turnout coefficient negative and significant, all columns sign-consistent: radio’s pro-Democratic, turnout-depressing effect strengthens in the FDR era, with a near-zero or mildly negative pre-FDR effect. I read the split cautiously: the design estimates radio’s average interwar

effect, and decomposing by era strains the identifying variation. Descriptive year-by-year effects (Appendix Table A8) shift toward the FDR era under the primary specification, while the cross-state pattern under county-and-year fixed effects peaks at 1928.

The 1928 peak under the cross-state specification has a ready historical reading unconnected to information poverty: Al Smith’s Catholicism and the Prohibition cleavage moved Catholic, immigrant, and wet counties sharply Democratic (Lichtman, 1979). But state-by-year fixed effects absorb the statewide religious wave, the spike appears only under the non-primary county-and-year specification, and the decomposition runs the other way: the Smith surge raised Democratic counts, whereas my within-state estimates show flat Democratic and declining Republican counts. A direct horse race sharpens the distinction (Appendix Table A19): instrumenting radio and both interactions jointly, the foreign-born interaction is robust ($+0.054$ per SD, $p < 0.01$) while the illiteracy interaction attenuates ($+0.019$, $p \approx 0.13$) on vote share, but the turnout gradients diverge in sign (illiteracy -0.049 , $p < 0.001$; foreign-born $+0.041$, $p < 0.05$): radio *mobilized* immigrant electorates (Andersen, 1979; Zonszein, 2025) while *demobilizing* habitual Republicans where illiteracy was high. Because the Census counts foreign-language literates as literate, the foreign-born margin partly proxies the same English-print exclusion.

This temporal pattern alone does not rule out FDR-specific messaging (Table 1): a mechanism operating only through Roosevelt-era broadcasts would predict exactly this split. What rules it out is the conjunction with the literacy gradient: if radio’s effect ran through FDR’s communication, there is no reason for it to scale with county illiteracy, since FDR’s audio reached literate and illiterate listeners alike. The illiteracy interaction therefore identifies information poverty, not FDR-specific messaging, as the operative channel, even granted the FDR era was when it became most consequential. Two controls corroborate this: county New Deal expenditure per capita (Fishback et al., 2003) interacted with a post-1932 indicator changes the radio $\times$ illiteracy coefficient by less than 5% (Appendix Table A10), and farm-distress controls — farmland share $\times$ year and

drought/Dust Bowl exposure $\times$ FDR era — move it by at most 16% with significance retained (Appendix Table A11).

Newspaper Circulation Within newspaper-tracked counties, the framework predicts that *literacy*, not newspaper availability, binds: the illiteracy gradient should replicate, while the gradient by circulation per capita should be weaker, since conditional on having a tracked daily, circulation moves the environment along a dimension voters already have the literacy to access. I test both using county circulation per capita (Gentzkow et al., 2011) in tracked counties; the test isolates the *intensive* margin: whether, among counties already served by print, more circulation further attenuates radio’s effect. It is silent by construction on the extensive margin, a fixed county trait the design differences out. The primary estimator is non-South IV with three endogenous regressors (radio, radio $\times$ illiteracy, radio $\times$ circulation per capita), instrumented by the six corresponding geographic-instrument interactions.

Table 8 replicates Table 6’s illiteracy-gradient test in the newspaper-tracked subsample and confirms the gradient: the radio level effects are sign-consistent with the main specification, and first-stage F-statistics are healthy throughout.

Two features stand out. First, the asymmetry between the two moderators is sharp and consistent across estimators: in the non-South, the radio $\times$ illiteracy interaction is significant under both OLS and IV, while the radio $\times$ circulation interaction is indistinguishable from zero under both. Because both are measured in the same sample under the same specification, this is not a difference in power or identification. Appendix Table A20 extends the test to the other two newspaper measures: the illiteracy interaction is unchanged (0.08–0.09 per SD), while paper count and total circulation carry small positive interactions about one-eighth the illiteracy gradient. Within the newspaper-tracked sample, the binding constraint radio relaxes is primarily *literacy*, not newspaper density.

Second, the turnout interactions clarify a sign pattern OLS alone would misread: the

Table 8: Newspaper Circulation, Illiteracy, and Radio’s Electoral Effects

	Non-South IV (primary)		Full-sample IV (supp.)		Non-South OLS (supp.)	
	Pres Dem Share	Pres Turnout	Pres Dem Share	Pres Turnout	Pres Dem Share	Pres Turnout
Radio	0.648*** (0.192)	−0.350** (0.119)	0.305* (0.154)	−0.221+ (0.132)	0.289*** (0.063)	0.031 (0.050)
Radio × Illit. (std.)	0.093* (0.037)	−0.004 (0.019)	0.036 (0.030)	0.002 (0.025)	0.079** (0.027)	0.026* (0.011)
Radio × Circ. p.c. (std.)	−0.001 (0.012)	0.022+ (0.012)	0.005 (0.006)	0.008 (0.007)	0.022 (0.016)	0.019* (0.009)
First-stage F (radio)	80	80	143	144	—	—
First-stage F (radio × illit)	748	747	822	820	—	—
First-stage F (radio × circ)	1,138	1,137	1,653	1,530	—	—
County FE	Yes	Yes	Yes	Yes	Yes	Yes
State × Year FE	Yes	Yes	Yes	Yes	Yes	Yes
N	4,418	4,412	5,548	5,539	4,424	4,418

Note: IV columns instrument the three endogenous radio terms with the two geographic instruments and their interactions with standardized illiteracy and standardized per-capita circulation (six instruments). OLS columns report the analogous specification without instrumenting. County and state-by-year fixed effects; illiteracy and per-capita circulation standardized within the newspaper-tracked sample (counties with tracked coverage, (Gentzkow et al., 2011)). Non-South = Confederate-11 (see Table 2 note). State-clustered SEs. + $p < 0.10$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

OLS radio × illiteracy turnout interaction is positive and significant here, while the IV counterpart is essentially zero, and in the full non-South sample the IV turnout interaction is significantly negative (Table 6). The preferred reading is that radio’s turnout response along the illiteracy dimension is weak or negative, not positive.⁸

That literacy binds while newspaper density does not is the test that separates information poverty from salience mobilization, which predicts symmetric positive moderation by both literacy and newspaper density (Table 1). Persuasion of print-informed audiences and FDR-specific messaging predict neither moderator to operate; elite signaling is adjudicated by the bipartisan-response evidence instead (Section 4.2). Among the voter-side mechanisms in Table 1, the joint pattern, positive literacy moderation with a null newspaper-density moderator in the same sample under the same specification, is

⁸The non-South IV illiteracy interaction slightly exceeds its OLS counterpart, the same IV > OLS ordering that appears at the level and at the Table 6 interaction.

predicted by information poverty alone.

The argument rests on conjunction. No single result is decisive; each is consistent with more than one mechanism in Table 1. But jointly only information-poverty reduction matches all of them: the illiteracy interaction rules out persuasion of print-informed audiences and strict FDR-specific messaging; the moderator asymmetry rules out salience mobilization; the conjunction of that literacy gradient with the FDR-era concentration, reinforced by the New Deal and farm-distress controls, closes off the milder Roosevelt-era-only version; the asymmetric Republican decline rules out entertainment crowd-out (Section 4.1); and the bipartisan legislator response rules out elite signaling. Radio shifted electoral outcomes by bringing political information to voters the print environment had excluded.

6 Robustness

First-stage strength is far above conventional weak-instrument thresholds throughout. State-by-year fixed effects absorb 59–63% of instrument variation at the county level (Table A2) and 79–84% at the district level, so the attenuation between specifications reflects removal of between-state confounding, not power loss.

Results are stable across additional specifications (Appendix Tables A7–A5): state-specific linear time trends added to the county-and-year specification leave the presidential vote-share estimate close to the state-by-year baseline; OLS estimates are positive, significant, and smaller than IV; the conductivity-only estimate attenuates from 0.250 to 0.154 while the decomposition and turnout results are essentially unchanged when woodland is dropped (Appendix Table A6); the balanced-panel subsample reproduces the coefficient almost exactly (0.251 versus 0.250); and farm-distress controls leave the headline essentially unchanged (Appendix Table A11). Among illiteracy, urbanization, and schooling as competing moderators of the district-level ideology effect, only illiteracy

retains significance (Appendix Table A18), a significance horse race whose sign instability parallels the county-level interaction-versus-split tension and locates in the linear-in-moderator functional form (Appendix E.1).

Two further diagnostics support the design. Hansen J tests of joint validity do not reject in any primary state-by-year specification (p between 0.14 and 0.80), rejecting only for the non-South county-and-year specification ($p = 0.002$), the same between-state variation the placebo implicates (Appendix Table A3). Weak-instrument-robust Anderson–Rubin sets for the headline non-South share effect exclude zero under both clusterings (Appendix Table A9).

Pre-Radio Placebo I estimate reduced-form regressions of the instruments on 1908–1916 presidential outcomes, before commercial broadcasting (Appendix Table A4). Ground conductivity $\times$ time is at most weakly associated with pre-radio Democratic vote share: under the state clustering appropriate to these spatially smooth instruments it is insignificant under both fixed-effect structures (non-South $p = 0.23$ to 0.36), reaching marginal significance only under county clustering ($p \approx 0.03$), and then negative. Woodland $\times$ time predicts pre-radio Democratic share under county-and-year fixed effects ($p < 0.001$), but state-by-year fixed effects absorb it ($p = 0.69$), indicating between-state geography rather than a within-state pre-trend. The marginal pre-trend that remains works *against* the headline: the pre-radio coefficients are negative, opposite the positive treatment-era reduced form, so extrapolating would attenuate not inflate the estimate. And the decomposition and turnout results survive dropping woodland (Appendix Table A6).

Leave-One-Out by State and Year I refit the headline non-South specification (Table 2, Panel A, column 2), dropping each of the 37 non-South states in turn and then each of the six presidential years. All 43 fits remain highly significant ($p < 0.001$); no exclusion flips the sign or removes significance. Dropping a state moves the coefficient

only modestly (0.215 to 0.365 around the headline 0.250); larger swings come from dropping a year (0.165 to 0.623), which reweights the within-county time variation. The shifts that strengthen the estimate are expected: excluding Kentucky, the high-leverage Border South cluster, and the no-radio 1920 baseline. The largest downward shifts stay well above conventional thresholds.

Subsample Heterogeneity by Region The headline averages over three structurally distinct subsamples. Splitting by Jim Crow severity — Northern Core (no Jim Crow), Border South (slave states that did not secede), and Confederate 11 — and refitting the headline IV on each yields three distinct estimates (Table A21): the Northern Core effect is roughly 2.2 times the headline, the Border South positive and significant but about 45% of it, and the Confederate 11 statistically indistinguishable from zero, consistent with the scope condition. The headline is thus a weighted average, with the Northern Core driving the bulk. The operative scope condition remains the South/non-South distinction, an institutional cliff, not a demographic gradient: the South was the most illiterate region, so a continuous illiteracy moderator would predict the largest effect there; instead it is null, because one-party Democratic rule left too few Republican voters to disrupt.

7 Conclusion

A new information technology rarely reaches all citizens equally, and its political consequences, for electoral outcomes and representation alike, depend on which population is newly reached and what information it encounters. I examined this through radio in interwar America, where audio bypassed the literacy barrier that had excluded millions from print-based political news.

Radio raised Democratic presidential vote share where the literacy barrier was highest, and it did so by eroding an inherited Republican allegiance rather than manufacturing new Democratic enthusiasm. The same disruption reached elected representatives, mov-

ing both parties leftward in high-illiteracy districts through distinct channels: Democratic adaptation as well as turnover, Republican turnover alone. That bipartisan, dual-channel response rules out single-party elite signaling and is the most demanding evidence for the voter-to-representative link in the accountability chain.

The condition for these effects is an uneven information environment: a new medium must reach a population the existing media had bypassed. In interwar America that dimension was literacy — information poverty. When the new medium reaches the excluded population, the resulting shock can disrupt habitual voting that information poverty had sustained. Radio disrupted a Republican partisanship inherited from the 1860s and re-economized by the 1896 realignment, producing asymmetric partisan effects whose direction depends on the content of the new information and the economic interests of the newly reached. The mechanism also requires a scope condition, a sufficiently large pool of misaligned voters whose behavior can shift. In the interwar South, one-party Democratic dominance left too few Republican voters for detectable aggregate effects, even though radio reached Southern households and literacy barriers were severe.

The framework speaks to contemporary media, with one refinement: the dimension of unevenness follows from each technology's distinctive affordance. Radio's audio format made *access* — who could reach political news — the binding dimension; some access gaps persist (broadband, news deserts), as Norris (2001) documents for the internet. Subsequent media shifted the dominant axis: television's synchronous video added unevenness in who *responded*, mobilizing voters who weighed image and affect over text; social media's networked, algorithmically curated diffusion moved it into what each audience *encountered*; and personalized large language models push further toward individualized information environments. Every medium has access, response, and content components, but the dominant axis differs by affordance, and matching each technology's affordance to the population it differentially reaches is where the framework's predictive content lies.

Several limitations bound the interpretation. My county-level design identifies ag-

gregate effects, not individual vote choices: the decomposition is consistent with some Republicans switching and others withdrawing, but I cannot observe individual transitions. I observe set-ownership shares rather than listening, so selection into ownership among the poor, mitigated but not eliminated by communal listening and the 1930s collapse in set prices, remains unobservable. And the findings are specific to one country, era, and medium.

What remains constant is the core insight: the political consequences of a new information technology depend on whom it reaches and the information environment they already inhabit, not on its content alone. When a new medium breaks through the barriers that sustained an uneven information environment, it does not merely add a voice to the conversation — it disrupts the political equilibrium that unevenness itself helped maintain.

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Appendix

A Data and Summary Statistics

Table A1 reports means and standard deviations for every variable entering the analysis, separately for the South and non-South.

B Instrument Validity and the First Stage

Identification rests on the residual within-state variation in the two geographic instruments. The tables in this section show that this residual is a small share of the raw instrument variance yet still yields a strong first stage (Table A2), that the per-instrument first-stage coefficients and the overidentification tests behave as expected (Table A3), and that the within-state design removes the instruments' association with pre-radio (1908–1916) presidential Democratic vote share in the non-South under the state-clustered inference appropriate to these instruments, with the one residual exception being woodland share's association with non-South turnout (Table A4).

C Robustness of the Headline Estimate

The non-South presidential vote-share effect is stable across the choices that most plausibly threaten it: using both within- and between-state variation rather than only within-state variation (Table A5), dropping the woodland instrument (Table A6), substituting milder state-specific time trends for saturated state-by-year effects (Table A7), decomposing the effect election-by-election (Table A8), clustering at the state rather than the county level (Table A9), and controlling for New Deal spending (Table A10) and farm distress (Table A11).

Table A1: Summary Statistics

Variable	N	Mean	SD	Min	Max
<i>Full sample</i>					
Pres. Dem vote share	18,322	0.565	0.235	0.033	1.000
Pres. turnout (fraction)	18,286	0.534	0.244	0.000	1.000
Radio ownership (fraction)	18,322	0.311	0.280	0.000	0.985
Illiteracy rate	18,322	0.045	0.049	0.000	0.441
Urbanization	18,322	0.210	0.250	0.000	1.000
Black population share	18,322	0.111	0.186	0.000	0.908
Log population	18,322	9.821	1.030	4.205	15.218
Ground conductivity	18,322	9.166	8.062	0.500	30.000
Woodland share	18,322	0.062	0.078	0.000	0.424
<i>Non-South</i>					
Pres. Dem vote share	11,693	0.451	0.171	0.033	1.000
Pres. turnout (fraction)	11,677	0.689	0.118	0.196	1.000
Radio ownership (fraction)	11,693	0.381	0.294	0.000	0.985
Illiteracy rate	11,693	0.023	0.026	0.000	0.397
Urbanization	11,693	0.242	0.259	0.000	1.000
Black population share	11,693	0.019	0.045	0.000	0.491
Log population	11,693	9.860	1.102	5.485	15.218
Ground conductivity	11,693	10.468	8.276	0.500	30.000
Woodland share	11,693	0.034	0.052	0.000	0.383
<i>South</i>					
Pres. Dem vote share	6,629	0.765	0.197	0.060	1.000
Pres. turnout (fraction)	6,609	0.260	0.150	0.000	1.000
Radio ownership (fraction)	6,629	0.188	0.201	0.000	0.965
Illiteracy rate	6,629	0.084	0.054	0.000	0.441
Urbanization	6,629	0.153	0.223	0.000	1.000
Black population share	6,629	0.275	0.223	0.000	0.908
Log population	6,629	9.752	0.885	4.205	13.179
Ground conductivity	6,629	6.869	7.111	1.000	30.000
Woodland share	6,629	0.110	0.091	0.000	0.424

Note: Canonical presidential estimation sample (1920–1940; county-years with non-missing outcome, radio, both instruments, and controls), closely matching the regression tables: N here is the raw count of county-years in the estimation sample, one to five observations larger per cell than the regression N (18,317 / 11,689 / 6,628 for vote share; 18,281 / 11,673 / 6,608 for turnout), which additionally drops fixed-effect singletons. Turnout is capped at 1.0 (39 county-years; see footnote 3 in the main text), so the values describe the variable as used in estimation. Radio ownership comes from the 1930 and 1940 Censuses (the first to ask about radio sets) and is linearly interpolated; the 1920 anchor is zero, as commercial broadcasting began only in November 1920. Illiteracy rate is time-varying (1920 Census through the 1924 election; 1930 Census from 1926 onward). South is the eleven former Confederate states. Ground conductivity and woodland share are time-invariant geographic instruments from Stromberg (2004); one county carried a source woodland-share value exceeding 1 (a coding artifact in the inherited replication data), which we set to missing, so it does not enter any woodland-instrumented specification.

Table A2: Instrument Variance Decomposition

Instrument	Var (Cty+Yr resid)	Var (St×Yr resid)	Marginal absorbed	Total R^2 (St×Yr)
Ground conductivity × time	2,891	1,197	59%	0.91
Woodland share × time	0.303	0.114	63%	0.92

Note: Variance of instrument residuals after partialling out fixed effects (House election sample). “Marginal absorbed” is the fraction of post-(county-and-year) residual variance additionally absorbed by state-by-year FE. “Total R^2 (St×Yr)” is the fraction of raw instrument variance absorbed by county and state-by-year FE jointly — only the residual 8–9% drives identification, yet first-stage F-statistics remain above 600.

Table A3: First-Stage Coefficients, Instrument Strength, and Overidentification

	Full		Non-South		South	
	S×Y	C+Y	S×Y	C+Y	S×Y	C+Y
Conductivity × time ($\times 10^3$)	0.174*** (0.017)	0.000 (0.018)	0.145*** (0.017)	−0.064*** (0.016)	0.206*** (0.040)	0.254*** (0.037)
Woodland × time	−0.029*** (0.002)	−0.062*** (0.002)	−0.056*** (0.003)	−0.057*** (0.003)	−0.013*** (0.003)	−0.003 (0.003)
Partial R^2 : conductivity	0.033	0.042	0.040	0.006	0.025	0.060
Partial R^2 : woodland	0.060	0.198	0.153	0.118	0.017	0.023
Conventional first-stage F	685	1,885	1,000	671	109	178
Kleibergen–Paap rk Wald F	147	565	274	138	25	44
Hansen J p -value	0.847	0.846	0.24	0.002	0.219	0.554
N	18,317	18,317	11,689	11,689	6,628	6,628

Note: First-stage regressions of interpolated radio ownership on the two excluded instruments, with controls and the indicated fixed effects, on the presidential vote-share sample. SE clustered at the county level; conductivity coefficients and SEs scaled by 10^3 . Partial R^2 is each instrument’s incremental within-FE R^2 over the controls-only first stage. “Conventional first-stage F” is the homoskedastic F-test of the excluded instruments (the statistic in the body tables); the Kleibergen–Paap row is its county-cluster-robust Wald equivalent. The Hansen J row reports two-step GMM overidentification tests (county-clustered): joint validity of the two instruments is not rejected in any state-by-year specification, and is rejected only in the non-South county-and-year cell ($p = 0.002$) — the same between-state variation flagged by the pre-radio placebo (Table A4) — consistent with the within-state design as primary; for the other primary outcomes under state-by-year FE it does not reject (p between 0.14 and 0.85). Because interpolated radio ownership carries measurement error, the second-stage IV estimates are attenuated and best read as a lower bound. ⁺ $p < 0.10$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

D Vote-Count Decomposition and the Demobilization Mechanism

The share-level effect decomposes into a sharp decline in Republican vote counts with no offsetting Democratic gain. This section states the accounting identity behind that pattern and reports the decomposition across fixed-effect structures, baseline partisan competi-

Table A4: Pre-Radio Placebo: Reduced-Form Instrument Effects, 1908–1916

	Dem Share		Turnout	
	Full	Non-South	Full	Non-South
<i>Panel A: County and Year FE</i>				
Conductivity $\times$ time	−0.000065 (0.000027)* [0.000067]	−0.000061 (0.000029)* [0.000066]	−0.000184 (0.000042)*** [0.000137]	−0.000259 (0.000049)*** [0.000137] ⁺
Woodland $\times$ time	−0.001 (0.004) [0.015]	−0.030 (0.004)*** [0.009]***	0.006 (0.004) [0.013]	−0.012 (0.006)* [0.012]
<i>Panel B: County and State $\times$ Year FE</i>				
Conductivity $\times$ time	−0.000093 (0.000035)** [0.000050] ⁺	−0.000086 (0.000040)* [0.000071]	0.000015 (0.000054) [0.000072]	−0.000034 (0.000069) [0.000104]
Woodland $\times$ time	0.007 (0.003)* [0.005]	−0.003 (0.003) [0.006]	−0.010 (0.007) [0.012]	−0.033 (0.017) ⁺ [0.016]*
Controls	Yes	Yes	Yes	Yes
County FE	Yes	Yes	Yes	Yes
N	8,744	5,579	8,566	5,437

Note: Reduced-form regressions of pre-radio outcomes (1908, 1912, 1916 presidential elections) on the instruments interacted with a linear time trend anchored at 1908; because the instruments are time-invariant, county fixed effects render the coefficients invariant to the anchor year. Controls as in Table 2, interacted with time. Each coefficient carries two SEs — county-clustered in parentheses, state-clustered in brackets, stars attached separately. Because the instruments are spatially smooth geographic fields $\times$ time, state clustering is the appropriate inference (see Table A9 note); county clustering is shown for comparison. Panel A uses county and year FE; Panel B replaces year with state-by-year FE. Under state clustering, conductivity $\times$ time passes the non-South placebos with a negative (headline-opposing) sign; woodland $\times$ time passes the non-South Democratic-share placebo under state-by-year FE but fails the non-South turnout placebo there (state $p = 0.04$), a limitation the conductivity-only estimates in Table A6 address. Democratic-share and turnout Ns differ because turnout is missing for some county-years. ⁺ $p < 0.10$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

tion, and county urbanization.

D.1 A Formal Accounting Framework

Each voting-age citizen has a type $\theta \in \{L, R\}$ (true policy alignment) and information $\sigma \in \{0, 1\}$ (whether they know their type). The informed ($\sigma = 1$) vote their type; the uninformed ($\sigma = 0$) vote by habit, independent of type: fractions v_D , v_R , and $1 - v_D - v_R$ vote Democratic, Republican, and abstain, respectively. Let P_U denote the uninformed

Table A5: County-and-Year Fixed Effects: Vote Share and Turnout

	(1) Full	(2) Non-South	(3) South
<i>Panel A: Presidential Dem vote share</i>			
Radio	0.452*** (0.024)	0.995*** (0.078)	0.289** (0.088)
First-stage F	1,885	671	178
N	18,317	11,689	6,628
<i>Panel B: House Dem vote share</i>			
Radio	0.182*** (0.029)	0.647*** (0.074)	0.025 (0.101)
First-stage F	3,211	1,204	367
N	34,315	21,717	12,598
<i>Panel C: Presidential turnout (fraction)</i>			
Radio	0.319*** (0.019)	0.293*** (0.045)	-0.536*** (0.103)
First-stage F	1,883	671	178
N	18,281	11,673	6,608
<i>Panel D: House turnout (fraction)</i>			
Radio	0.311*** (0.013)	0.271*** (0.032)	0.330*** (0.060)
First-stage F	3,261	1,136	366
N	34,904	22,321	12,583
County FE	Yes	Yes	Yes
Year FE	Yes	Yes	Yes

Note: IV estimates with county and year fixed effects, using both within- and between-state variation in radio exposure. Controls, instruments, and clustering as in Table 2, but with year (not state-by-year) FE. The substantially larger coefficients relative to the state-by-year specification (Tables 2 and 3) reflect between-state confounding: favorable-reception states were also more developed, urbanized, and competitive. The positive turnout estimates (Panels C and D) reverse under state-by-year FE, indicating the baseline turnout effect is between-state, not causal. Panel C turnout is capped at 1.0 as in all turnout regressions; Panel D is total House votes divided by county population. Ns differ by one or two observations across FE structures because singleton cells are dropped. ⁺ $p < 0.10$, $*p < 0.05$, $**p < 0.01$, $***p < 0.001$.

population and α the fraction of type- L citizens within it. In the interwar non-South the uninformed were disproportionately type- L , so $\alpha > 0.5$, and habitual voting leaned Republican, so $v_R \geq v_D$.

With these primitives, the key misaligned groups are type- L citizens voting Republican by habit, $R_L = \alpha \cdot v_R \cdot P_U$, and type- R citizens voting Democratic by habit, $D_R = (1 - \alpha) \cdot v_D \cdot P_U$. Because $\alpha > 0.5$ and $v_R \geq v_D$, it follows that $R_L > D_R$.

Table A6: Single-Instrument IV Estimates: Ground Conductivity Only

	Combined IV		Conductivity Only		
	Coef (SE)	F-stat	Coef (SE)	F-stat	N
<i>Presidential Dem Share</i>					
Full, County+Year	0.452*** (0.024)	1,885	0.441*** (0.058)	672	18,317
Full, State×Year	0.092 ⁺ (0.051)	685	0.080 (0.081)	511	18,317
Non-South, County+Year	0.995*** (0.078)	671	1.834*** (0.358)	61	11,689
Non-South, State×Year	0.250*** (0.040)	1,000	0.154 (0.094)	392	11,689
<i>Presidential Turnout</i>					
Non-South, State×Year	−0.322*** (0.037)	999	−0.306*** (0.070)	392	11,673
<i>Vote Decomposition (Non-South, State×Year)</i>					
Log Dem	0.039 (0.170)	1,000	−0.377 (0.383)	392	11,689
Log Rep	−1.069*** (0.116)	1,000	−1.476*** (0.320)	392	11,689

Note: IV estimates comparing the combined-instrument specification (conductivity × time and woodland × time) with conductivity × time alone, both on the common canonical sample requiring both instruments non-missing, so Ns match the main tables. SE clustered at county level; conventional first-stage F on the excluded instrument(s). The core findings survive dropping woodland: the Republican-count and turnout declines are similar across columns. The presidential Dem share estimate under state-by-year FE attenuates from 0.250 to 0.154 and loses significance with the single instrument; the conductivity-only interval is consistent with the combined estimate but does not establish the headline share effect on its own. First-stage F exceeds conventional thresholds throughout (weakest 61, non-South county-and-year). * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

Table A7: Robustness: State-Specific Linear Time Trends

	County + Year	State Trends	State × Year
<i>Panel A: Full sample</i>			
Pres Dem Share	0.452***	0.121*	0.092 ⁺
House Dem Share	0.182***	0.101 ⁺	0.103 ⁺
<i>Panel B: Non-South</i>			
Pres Dem Share	0.995***	0.256***	0.250***
House Dem Share	0.647***	0.196***	0.181**

Note: IV estimates under county clustering. The State Trends column adds state-specific linear time trends to the county-and-year specification; the rightmost column instead uses saturated state-by-year fixed effects. The milder state-trend control reproduces the state-by-year estimate almost exactly in the non-South (0.256 vs 0.250 for the presidential effect) and lands close to it in the full sample (0.121 vs 0.092), indicating that the attenuation from county-and-year to state-by-year fixed effects reflects the removal of trending between-state confounders rather than power loss. ⁺ $p < 0.10$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

Radio reveals θ to uninformed citizens, inducing six flows: mobilization of left non-voters (*a*) and right non-voters (*b*); demobilization of misaligned Democrats (*c*) and misaligned Republicans (*e*); and switching by misaligned Democrats to Republican (*d*) and

Table A8: Year-by-Year OLS Estimates: Radio and Presidential Dem Share

Election Year	Cumulative Radio Coef	President
1928	0.449	Hoover (R)
1932	0.303	FDR (D)
1936	0.370	FDR (D)
1940	0.173	FDR (D)

Note: OLS with county and year fixed effects. Each row is the within-county radio effect on presidential two-party Democratic vote share in that election year (cumulative interpolated radio ownership $\times$ election-year indicators). The 1924 election is omitted because radio ownership was still near zero. Full sample ($N = 18,321$, estimated on the common instrument sample so N matches the IV tables). Under county and year FE (which retain between-state variation) the strongest single-year effect is in 1928; under state-by-year FE the within-state pattern shifts toward the FDR era (Table 7), illustrating the sensitivity of temporal decomposition to specification. $^*p < 0.05$, $^{**}p < 0.01$, $^{***}p < 0.001$.

Table A9: Headline Level Estimates Under County- and State-Clustered Standard Errors

Outcome	Sample	Coef	SE (county)	SE (state)
Pres Dem share	Full	0.092	(0.051) ⁺	(0.112)
	Non-South	0.250	(0.040) ^{***}	(0.103) [*]
	South	-0.109	(0.132)	(0.317)
House Dem share	Full	0.103	(0.061) ⁺	(0.112)
	Non-South	0.181	(0.056) ^{**}	(0.100) ⁺
	South	-0.020	(0.121)	(0.169)
Pres turnout	Full	-0.125	(0.042) ^{**}	(0.105)
	Non-South	-0.322	(0.037) ^{***}	(0.062) ^{***}
	South	0.192	(0.109) ⁺	(0.149)
House turnout	Full	-0.032	(0.029)	(0.078)
	Non-South	-0.173	(0.034) ^{***}	(0.078) [*]
	South	0.138	(0.060) [*]	(0.086)
Log Dem votes	Non-South	0.039	(0.170)	(0.526)
Log Rep votes	Non-South	-1.069	(0.116) ^{***}	(0.138) ^{***}
Log Total votes	Non-South	-0.438	(0.071) ^{***}	(0.181) [*]

Note: Every cell re-estimates the corresponding state-by-year IV level specification from Tables 2–4 and reports both clustering levels; stars attach to each SE separately. Because the instruments are spatially smooth geographic fields $\times$ time, state clustering is the more conservative inference. The substantive core survives it: the non-South presidential share effect stays significant at 5%, and the turnout and Republican-count declines stay significant at 0.1%. The full-sample share estimates — supplementary throughout, since the scope condition makes the non-South operative — lose significance under state clustering. Weak-instrument-robust Anderson–Rubin 95% confidence sets for the non-South share effect are $[0.17, 0.34]$ (county) and $[0.01, 0.51]$ (state), both excluding zero. $^+p < 0.10$, $^*p < 0.05$, $^{**}p < 0.01$, $^{***}p < 0.001$.

misaligned Republicans to Democratic (f).

Table A10: Robustness: Illiteracy Gradient Controlling for New Deal Spending

	(1) Baseline (full)	(2) Fishback subsample	(3) + Total NDEXP	(4) + Relief + AAA
Radio	0.291** (0.107)	0.278** (0.104)	0.272** (0.103)	0.289** (0.101)
Radio $\times$ Illiteracy	1.620* (0.635)	1.538* (0.624)	1.566* (0.623)	1.632** (0.606)
NDEXP per capita $\times$ FDR era			6.5×10^{-5} ** (2.3×10^{-5})	
County FE	Yes	Yes	Yes	Yes
State $\times$ Year FE	Yes	Yes	Yes	Yes
N	11,689	11,653	11,653	11,653

Note: IV estimates, non-South sample, county and state-by-year fixed effects. The endogenous variables (radio and radio $\times$ illiteracy) are instrumented by ground conductivity $\times$ time, woodland $\times$ time, and their interactions with county illiteracy; SE clustered at the state level. Column (1) replicates the non-South specification of Table 6, Panel A, column (1) on the canonical sample. Column (2) restricts to the Fishback et al. (2003) New Deal expenditure subsample (1,950 of 1,956 non-South counties matched); columns (3)–(4) add total New Deal expenditure per capita $\times$ post-1932, and relief and AAA spending $\times$ post-1932 separately. The radio $\times$ illiteracy coefficient moves by at most about 5% across specifications, so the illiteracy gradient is not attributable to FDR-era policy targeting of high-illiteracy counties; the spending terms themselves predict Democratic vote share, so the design controls for rather than ignores the returns to New Deal spending. ⁺ $p < 0.10$, $*p < 0.05$, $**p < 0.01$, $***p < 0.001$.

Table A11: Robustness: Farm-Distress Controls

	(1) Baseline (full)	(2) Fishback subsample	(3) + Farm share × year FE	(4) + Drought, Dust Bowl × FDR	(5) + All
<i>Panel A: Headline level effect (Pres Dem share)</i>					
Radio	0.250	0.237	0.311	0.239	0.308
SE (county)	(0.040)***	(0.040)***	(0.042)***	(0.040)***	(0.042)***
SE (state)	(0.103)*	(0.100)*	(0.104)**	(0.102)*	(0.104)**
First-stage F	1,000	997	924	984	923
<i>Panel B: Illiteracy interaction (state-clustered)</i>					
Radio	0.291** (0.107)	0.278** (0.104)	0.335** (0.106)	0.280** (0.105)	0.334** (0.106)
Radio × Illiteracy	1.620* (0.635)	1.538* (0.624)	1.292* (0.542)	1.603* (0.640)	1.360* (0.550)
County FE	Yes	Yes	Yes	Yes	Yes
State × Year FE	Yes	Yes	Yes	Yes	Yes
N	11,689	11,653	11,653	11,653	11,653

Note: IV estimates, non-South sample (Confederate-11), county and state-by-year fixed effects. Farm-distress measures come from the Fishback et al. (2003) county files (farm-land share, 1930s drought months, a Dust Bowl indicator); 1,950 of 1,956 non-South counties match. Column (3) interacts standardized farm-land share with election-year FE; column (4) interacts drought and Dust Bowl with the FDR-era indicator; column (5) includes all. Panel A reports the single-endogenous headline specification with both clustering levels (stars attach to each SE separately); Panel B reports the two-endogenous illiteracy-interaction specification of Table 6 with state-clustered SE. The headline effect is essentially unchanged or slightly larger under these controls, and the illiteracy interaction moves by at most 16% relative to the matched baseline while remaining significant. These controls absorb farm *exposure* and 1930s drought shocks; a time-varying county farm-income measure would be needed to control distress directly and remains for future work. ⁺ $p < 0.10$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

The observable changes are:

$$\Delta D = a + f - c - d \quad (3)$$

$$\Delta R = b + d - e - f \quad (4)$$

$$\Delta T = a + b - c - e \quad (5)$$

The empirical finding $\Delta D \approx 0$ does not imply that no flows affected Democratic votes. Rather, it implies approximate cancellation: inflows (mobilization of left non-voters a and switching from misaligned Republicans f) roughly offset outflows (demobilization c and switching d of right-leaning habitual Democrats who discover their misalignment). The large $|\Delta R|$ reflects the structural asymmetry $R_L \gg D_R$: many more voters were misaligned on the Republican side.

Because the system has four degrees of freedom (six unknowns, two independent equations), individual flows cannot be identified from aggregate data. Three features do follow directly from the signs and magnitudes of the estimated aggregate changes: (1) total Republican exits (demobilization e plus switching f) exceed Republican-side inflows ($b + d$) by the magnitude of the Republican count decline, which is large; (2) net demobilization exceeds net mobilization ($c + e > a + b$), which is what the turnout decline expresses; and (3) Democratic-side inflows and outflows approximately cancel. The aggregate moments alone cannot apportion Republican exits between demobilization and switching. The auxiliary evidence on baseline competition (Appendix Table A14) constrains the solution further: detectable switching ($f > 0$) appears only in safe Republican counties where R_L is largest, while in competitive counties the Republican decline arrives without Democratic gains, pointing to demobilization as the dominant exit channel there.

Table A12: Vote Count Decomposition by Fixed Effects Specification (Non-South)

	Log Dem	Log Rep	Dem Share	Turnout
<i>Panel A: County and Year FE</i>				
Radio	3.577*** (0.273)	-2.229*** (0.348)	0.995*** (0.078)	0.293*** (0.045)
First-stage F	671	671	671	671
N	11,689	11,689	11,689	11,673
<i>Panel B: County and State $\times$ Year FE</i>				
Radio	0.039 (0.170)	-1.069*** (0.116)	0.250*** (0.040)	-0.322*** (0.037)
First-stage F	1,000	1,000	1,000	999
N	11,689	11,689	11,689	11,673
County FE	Yes	Yes	Yes	Yes

Note: IV estimates, non-South sample. Controls, instruments, and significance conventions as in Table 2. Standard errors clustered at county level in parentheses. Under county-and-year fixed effects, which use both between-state and within-state instrument variation, the positive turnout estimate is driven by large increases in Democratic votes across states (3.58) alongside a large Republican decline (-2.23). Under state-by-year fixed effects, which restrict identification to within-state variation, the pattern shifts: Democratic votes are unchanged while Republican votes decline sharply (-1.07), producing a net turnout decline. ⁺ $p < 0.10$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

D.2 Decomposition Estimates

The asymmetry between an essentially unchanged Democratic count and a sharply lower Republican count holds across fixed-effect structures (Table A12, with the South/non-South contrast in Table A13) and is concentrated in safe-Republican (Table A14) and, geographically, rural counties (Table A15).

Vote count decomposition by urbanization. The demobilization channel is concentrated in rural and small-town counties. Splitting the non-South at the median of baseline (1920) county urbanization and re-estimating the main IV decomposition in each half (Appendix Table A15), the rural half — the Republican farm belt of the Plains and border North (Kentucky, Missouri, Nebraska, Kansas, the Dakotas) — carries the demobilization signature: radio cut Republican votes, total votes, and turnout, each significant at $p < 0.001$, with no offsetting Democratic gain (the log Democratic coefficient is negative and insignificant at the median, and significantly negative in the bottom urbanization

Table A13: Vote Count Decomposition: South vs. Non-South (State $\times$ Year FE)

	Log Dem	Log Rep	Dem Share	Turnout
<i>Panel A: Non-South</i>				
Radio	0.039 (0.170)	-1.069*** (0.116)	0.250*** (0.040)	-0.322*** (0.037)
First-stage F	1,000	1,000	1,000	999
N	11,689	11,689	11,689	11,673
<i>Panel B: South</i>				
Radio	0.351 (0.375)	0.030 (0.827)	-0.109 (0.132)	0.192 ⁺ (0.109)
First-stage F	109	109	109	110
N	6,628	6,628	6,628	6,608
County FE	Yes	Yes	Yes	Yes
State $\times$ Year FE	Yes	Yes	Yes	Yes

Note: IV estimates with county and state-by-year fixed effects. Controls, instruments, and significance conventions as in Table 2. Standard errors clustered at county level in parentheses. In the non-South, the asymmetry between log Democratic votes (near zero) and log Republican votes (-1.07) drives the vote share increase and turnout decline. In the South, no log-vote component is statistically significant, consistent with the scope condition: too few Republican voters remained for the information shock to produce detectable aggregate shifts. ⁺ $p < 0.10$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

tercile). The urban half — the industrial states (Ohio, Illinois, New York, Pennsylvania, Michigan) — shows the opposite composition: a positive Democratic vote-count effect with no turnout decline, the mobilization pattern Andersen (1979) documents for the immigrant cities, though estimated imprecisely given the smaller number of urban counties and state clusters. The contrast is robust to tercile-extreme and majority-rural (urb < 0.5) splits. This within-state evidence locates the aggregate decomposition's two channels geographically: Republican demobilization on the rural margin, where habitual Republican loyalties were strongest, and a weaker Democratic mobilization in the cities.

D.3 The New Deal Realignment Debate

The vote-count decomposition speaks directly to one of the oldest questions about the New Deal realignment: whether the Democratic majority was built by converting Republicans or by mobilizing new voters (Key, 1955; Burnham, 1970; Andersen, 1979; Erik-

Table A14: Vote Count Decomposition by Baseline Competition (Non-South, State $\times$ Year FE)

	Safe Rep (<40%) $N_{\text{cty}} = 1,379$	Competitive (40–60%) $N_{\text{cty}} = 497$	Safe Dem (>60%) $N_{\text{cty}} = 66$
Log Dem	0.930* (0.445)	−0.273 (0.167)	−0.429* (0.187)
Log Rep	−1.522*** (0.262)	−0.838*** (0.172)	−0.195 (0.316)
Log Total	−0.172 (0.144)	−0.610*** (0.117)	−0.365* (0.178)
Dem Share	0.505*** (0.106)	0.136** (0.047)	−0.050 (0.060)
First-stage F	212	413	244
N	8,268	2,946	378
County FE	Yes	Yes	Yes
State $\times$ Year FE	Yes	Yes	Yes

Note: IV estimates, non-South sample (Confederate-11), split by 1920 baseline Democratic presidential vote share. Controls, instruments, and clustering as in Table 2; conventional first-stage F on the excluded instruments per column. Safe Republican counties show Democratic-vote gains alongside Republican declines, a pattern consistent with switching; competitive counties show Republican declines without Democratic gains, suggesting demobilization predominated. The Safe Dem column rests on only 66 counties and is reported for completeness; its modest negative log Democratic and log total coefficients are consistent with demobilization falling on habitual *Democratic* voters where Democrats dominated, the mirror image of the safe-Republican pattern. $^+p < 0.10$, $^*p < 0.05$, $^{**}p < 0.01$, $^{***}p < 0.001$.

son and Tedin, 1981; Sundquist, 1983). Andersen (1979) locates it in the mobilization of previously non-voting urban ethnics; Erikson and Tedin (1981) counter that conversion of former Republicans did much of the work. Both explain the Democratic majority as growth in the Democratic vote. My within-state estimates isolate a mechanism neither anticipates: a Democratic-share gain with no growth in Democratic votes at all, produced through the demobilization of habitual Republicans into abstention. Splitting the non-South by baseline urbanization locates this channel (Table A15): in the rural Republican farm belt radio sharply reduced Republican votes and depressed turnout with no offsetting Democratic gain — the demobilization signature, precisely estimated — while in the cities the same Republican decline was offset by rising Democratic votes, the mobilization Andersen (1979) describes, though imprecisely estimated here. Because state-by-year fixed effects absorb the national mobilization wave, my within-county estimates recover

Table A15: Vote-Count Decomposition by County Urbanization (Non-South, IV)

	Rural (low-urb)	Urban (high-urb)	All Non-South
Pres. Dem. share	0.125 (0.089)	0.397 ⁺ (0.214)	0.250* (0.103)
Log Dem. votes	-0.622 (0.400)	0.989 (0.952)	0.039 (0.526)
Log Rep. votes	-1.105*** (0.147)	-0.970*** (0.294)	-1.069*** (0.138)
Log total votes	-0.673*** (0.158)	-0.087 (0.347)	-0.438* (0.181)
Turnout	-0.389*** (0.062)	-0.177 (0.124)	-0.322*** (0.062)
County FE	Yes	Yes	Yes
State × Year FE	Yes	Yes	Yes
First-stage F	521	214	1,000
N	5,827	5,844	11,689

Note: Each cell is a separate single-endogenous IV estimate of the ratio coefficient, with county and state-by-year fixed effects and controls, non-South sample (Confederate-11). SE clustered at the state level. The sample is split at the non-South median of baseline (1920) county urbanization (0.17); the “All Non-South” column reproduces Table 4. The rural half displays the demobilization signature — a sharp Republican and turnout decline with no Democratic gain; the urban half shows the reverse, a Democratic vote gain without a turnout loss, but less precisely (weaker first stage, fewer state clusters). Tercile-extreme and majority-rural (urb < 0.5) splits give the same contrast, with the rural log-Democratic coefficient turning significantly negative (-0.80 , $p < 0.05$) in the bottom urbanization tercile. The turnout rows are estimated on a few county-years fewer than the column N shown, owing to the capped-denominator filter. ⁺ $p < 0.10$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

its two within-state margins: a rural Republican demobilization the debate never considered, and an imprecise echo of the urban mobilization it did.

The canonical synthesis did examine the countryside. Sundquist (1983, pp. 199–211) analyzes the rural North through the conversion lens and finds it “experienced no such conversion” as the cities, with a conservative rural countermovement offsetting urban Democratic gains — but identifies no media or information channel. Under the demobilization mechanism my decomposition recovers, a null in net rural realignment is exactly what such a lens should report. Darmofal (2008) likewise maps the rural realignment’s geography without its mechanism, leaving open the channel this paper isolates.

E The County Illiteracy Gradient

Radio’s pro-Democratic effect is steeper where county illiteracy is higher. This section reports the preferred continuous-interaction IV and its count-level decomposition, the less-informative subsample split and why it is less informative, the competing moderators that might rival illiteracy, and the regional pattern of the headline estimate.

E.1 Interaction IV versus Subsample Splits

My primary mechanism test for county illiteracy (Table 6) is a continuous interaction IV that treats radio and $\text{radio} \times \text{illiteracy}$ as endogenous, mirroring the main specification at both the level and the interaction. A less-efficient alternative is to split the non-South sample at the illiteracy median (or at the top-third cutoff) and estimate the main IV separately in each subsample.⁹

Sample composition: border-state counties disproportionately populate the high-illiteracy subsample. When I restrict the non-South sample (Confederate-11 definition) to counties above the top one-third illiteracy cutoff, Kentucky alone contributes 18% of the high-illiteracy observations. Kentucky is a border state whose post-Reconstruction politics combined Democratic dominance with Appalachian Republican enclaves, and its counties exhibit a ceiling-type vote count pattern. When Kentucky is excluded, the non-South top-third IV yields a Democratic vote share coefficient of $+0.39$ ($p < 0.01$, state-

⁹The median split (Non-South median $\text{illit}_{ct} \approx 0.014$) is used for the balanced IV subsample contrast in Appendix Table A17; the top-third cutoff (≈ 0.022) is used for the Kentucky-composition diagnostic below, where focusing on the marginal high-illiteracy cells makes the border-state ceiling pattern legible. Both definitions place the same counties on the same side of the cut for the bulk of the distribution. I document that alternative here for completeness and to show that the subsample-split results are broadly consistent with the interaction-IV evidence but exhibit sample-composition and instrument-heterogeneity features that make the subsample split a less informative test of the illiteracy gradient than the interaction IV is. As Appendix Table A17 shows, the Democratic vote share coefficient in the median split is larger in the low-illiteracy non-South subsample (0.707) than in the high-illiteracy subsample (0.217), an apparent reversal that dissolves once one accounts for the composition and instrument-heterogeneity issues documented below. Both subsample coefficients are nevertheless same-signed, so the split does not overturn the directional finding that radio increased Democratic vote share in the non-South. The interaction IV in Table 6 is the preferred test of the illiteracy gradient itself.

clustered), a log Democratic votes coefficient of +1.11 (positive but imprecise, $p > 0.10$), and a log Republican votes coefficient of -1.31 ($p < 0.05$), consistent with the mechanism's prediction. Further excluding Oklahoma and West Virginia — the other two non-South states with substantial Appalachian Republican pockets — leaves the Democratic vote share coefficient at +0.35 and the log Democratic votes coefficient at +1.10, qualitatively the same result. Because interaction IV pools across the full non-South and uses within-county over-time variation rather than cross-county subsample splits, it is not sensitive to these composition asymmetries in the same way.

Instrument heterogeneity: the two instruments identify partly different complier populations. When the IV is estimated using each instrument separately on subsample splits of the non-South, the two instruments give related but non-identical pictures of the illiteracy gradient: they generally agree in sign but place different weights on the complier populations, so any single subsample split reflects a weighted average of LATEs that need not coincide with the policy-relevant parameter. The continuous interaction IV in Table 6 is not fully immune to this issue, but it aggregates the gradient across the full non-South rather than from a narrow subsample contrast, reducing the sensitivity of the point estimate to instrument-specific weighting.

Vote count decomposition. Appendix Table A16 extends the IV illiteracy interaction to the vote count outcomes. The interaction with log Republican votes is negative and significant (-4.40 , $p < 0.001$), indicating that Republican vote decline is steeper in high-illiteracy counties; the interaction with log Democratic votes is same-signed (1.85) but too noisy under IV to identify a gradient. This pattern attributes the share-level illiteracy gradient (Table 6) primarily to differential Republican decline at high illiteracy, with Democratic-side flows insufficiently identified at the count level. The near-zero change in average non-South Democratic vote counts documented in Table 4 is consistent with the formal decomposition in Appendix D.1, where Democratic-side inflows and outflows can

Table A16: IV Radio $\times$ Illiteracy Interaction: Vote Count Decomposition (Non-South)

	Dem Share	Log Dem	Log Rep	Turnout
Radio	0.291** (0.107)	0.151 (0.545)	-1.116*** (0.118)	-0.327*** (0.060)
Radio $\times$ Illiteracy	1.620* (0.635)	1.853 (3.376)	-4.395*** (1.181)	-1.421** (0.542)
County FE	Yes	Yes	Yes	Yes
State $\times$ Year FE	Yes	Yes	Yes	Yes
N	11,689	11,689	11,689	11,673

Note: IV estimates with county and state-by-year fixed effects, non-South sample (Confederate-11). Standard errors (state-clustered) in parentheses. Both radio and radio $\times$ illiteracy are endogenous, instrumented by ground conductivity and woodland share interacted with time and with illiteracy. The Democratic vote share column replicates Table 6, column (1) of Panel A. The illiteracy gradient operates through the log Republican count (-4.40 , $p < 0.001$); the log Democratic count interaction is same-signed (1.85) but too imprecise to identify a gradient under IV. Per-equation first-stage F-statistics: radio ≈ 717 , radio $\times$ illiteracy $\approx 1,737$ (Pres Turnout marginally lower owing to a 16-observation drop from the capped-denominator filter); Sanderson–Windmeijer conditional F-statistics, the appropriate per-regressor diagnostic, are 162 (radio) and 73 (radio $\times$ illiteracy) under county clustering and 106 and 46 under state clustering — all well above weak-instrument thresholds. $^+p < 0.10$, $^*p < 0.05$, $^{**}p < 0.01$, $^{***}p < 0.001$.

cancel; the dominant identifiable count-level driver of the share gradient is the Republican channel.

I note one residual tension between estimators. The interaction-IV in Table A16 gives a log Republican coefficient of -4.40 ($p < 0.001$) on the radio $\times$ illiteracy term, implying steeper Republican decline at high illiteracy. The median-split IV in Table A17, however, shows a *larger* log Republican coefficient in absolute value in the *low*-illiteracy half (-2.73) than in the high-illiteracy half (-1.13). The share-level gradient is consistently positive across both estimators (high coefficient 0.217 , low 0.707 , both significant), but the count-level gradient on log Republican votes flips sign across the linear interaction and binary split, most likely reflecting a non-linear functional form that the two estimators summarize differently. I treat the interaction IV as the primary specification for the count-level gradient and note this disagreement as an open issue rather than a resolved finding.

Table A17: IV Vote Count Decomposition by County Illiteracy (Non-South, Median Split, State $\times$ Year FE)

	High Illiteracy	Low Illiteracy
Dem Share	0.217*** (0.045)	0.707** (0.273)
Log Dem Votes	−0.007 (0.200)	−0.257 (1.042)
Log Rep Votes	−1.127*** (0.165)	−2.732*** (0.572)
Turnout	−0.324*** (0.046)	−0.440** (0.157)
First-stage F	649	53
N	5,841	5,832
County FE	Yes	Yes
State $\times$ Year FE	Yes	Yes

Note: IV estimates, non-South sample (Confederate-11; see Table 2 note), split at the non-South median of time-varying county illiteracy. SE clustered at county level. The Democratic vote share coefficient is roughly three times larger in the low-illiteracy subsample than in the high-illiteracy subsample, the opposite of what the mechanism would predict if the split were a clean test of the gradient. Appendix E.1 documents why a subsample split is a less informative test of the literacy gradient than the interaction IV (sample composition, single-instrument heterogeneity). The continuous interaction IV in Table 6 — with the count decomposition in Table A16 — is the preferred test. The turnout row is estimated on a few county-years fewer than the column N shown, owing to the capped-denominator filter. $^+p < 0.10$, $^*p < 0.05$, $^{**}p < 0.01$, $^{***}p < 0.001$.

E.2 Competing Moderators

County illiteracy could proxy for other traits that radio plausibly activated. Three horse-race tests address this: Table A18 pits it against urbanization and schooling at the district level, Table A19 against county foreign-born share, and Table A20 against alternative newspaper-access measures.

E.3 Regional Heterogeneity

The headline non-South estimate is a weighted average across regions with very different baseline partisan structures. Table A21 refits the headline IV separately on the Northern Core, the Border South, and the Confederate-11 South.

Table A18: Competing Moderators (Horse Race)

	Nokken-Poole (1st dim.)
Radio $\times$ Illiteracy (std.)	0.261*** (0.050)
Radio $\times$ Urbanization (std.)	0.021 (0.036)
Radio $\times$ Schooling (std.)	-0.091 ⁺ (0.050)
District FE	Yes
Congress FE	Yes

Note: OLS with all three interactions simultaneously; district and Congress FE; SE clustered at district level. The dependent variable is the Nokken-Poole first-dimension score (negative = more liberal), so the positive Radio $\times$ Illiteracy coefficient implies a *weaker* leftward shift at high illiteracy — the opposite direction from the saturated-split IV gradient in Table 5, a district-level analogue of the interaction-versus-split sign instability documented in Appendix E.1. The instability is a functional-form artifact of the linear-in-moderator specification: re-estimating each moderator in the saturated-binary design of Table 5 yields a high-minus-low illiteracy difference of -1.10 (SE 0.26, $p < 0.001$) — aligned with Table 5’s gradient — while schooling runs in the expected opposite direction (+1.21, SE 0.27) and urbanization at +0.58 (SE 0.26). I therefore read this table as a significance horse race among competing moderators, not as directional evidence on the gradient. ⁺ $p < 0.10$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

Table A19: Horse Race: Illiteracy versus Foreign-Born Moderators

	Pres Dem Share			Pres Turnout		
	(1)	(2)	(3)	(4)	(5)	(6)
Radio	0.328** (0.119)	0.176*** (0.051)	0.210*** (0.060)	-0.360*** (0.069)	-0.342*** (0.049)	-0.400*** (0.052)
Radio $\times$ Illit. (std.)	0.043* (0.017)		0.019 (0.012)	-0.037** (0.014)		-0.049*** (0.013)
Radio $\times$ For.-born (std.)		0.054** (0.020)	0.054** (0.020)		0.026 (0.017)	0.041* (0.019)
County FE	Yes	Yes	Yes	Yes	Yes	Yes
State $\times$ Year FE	Yes	Yes	Yes	Yes	Yes	Yes
N	11,689	11,689	11,689	11,673	11,673	11,673

Note: Non-South IV with county and state-by-year fixed effects; SE clustered at the state level. Moderators are standardized within the estimation sample (raw illiteracy mean 0.023, SD 0.026; foreign-born mean 0.094, SD 0.098; within-sample correlation -0.01, so the horse race is not a collinearity contest). Every radio term is endogenous and instrumented by the corresponding interactions of the two geographic instruments (columns 3 and 6 use three endogenous regressors and six instruments; per-equation first-stage F 514, 1,517, 889). Level terms of each moderator enter as exogenous controls. In raw units the column (1) illiteracy interaction replicates the Table 6 estimate (1.620, SE 0.635). On vote share the foreign-born moderator survives the horse race while illiteracy attenuates ($p \approx 0.13$); on turnout the two diverge in sign, the signature of mobilization in immigrant counties versus demobilization in high-illiteracy counties. ⁺ $p < 0.10$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

Table A20: Newspaper Moderator Asymmetry: Alternative Newspaper Measures

	Moderator: paper count		Moderator: total circulation	
	Pres Dem Share	Pres Turnout	Pres Dem Share	Pres Turnout
Radio	0.567** (0.176)	−0.420*** (0.112)	0.650*** (0.196)	−0.373** (0.126)
Radio × Illit. (std.)	0.081* (0.034)	−0.019 (0.019)	0.089* (0.036)	−0.014 (0.020)
Radio × Newspaper (std.)	0.010** (0.003)	0.015*** (0.003)	0.013+ (0.007)	0.021*** (0.006)
First-stage F (radio)	84	84	85	85
First-stage F (radio × illit)	737	736	738	737
First-stage F (radio × news)	1,616	1,613	982	981
County FE	Yes	Yes	Yes	Yes
State × Year FE	Yes	Yes	Yes	Yes
N	4,418	4,412	4,418	4,412

Note: Non-South IV re-estimation of Table 8 replacing circulation per capita with the count of distinct tracked dailies and total daily circulation, each standardized within the tracked sample. Instruments and specification as in Table 8; SE clustered at the state level. The illiteracy interaction on Democratic vote share is stable across all three newspaper controls (0.081–0.093 per SD). The scale measures carry small positive vote-share interactions (0.010–0.013 per SD, roughly one-eighth the illiteracy gradient per SD), unlike the null per-capita measure; count and total circulation track market size and newspaper entry/exit rather than print-access intensity, so per-capita circulation remains the theoretically relevant access-intensity moderator. Within-county SDs of the standardized moderators are 0.13 (illiteracy), 0.15 (circulation per capita), 0.30 (count), and 0.18 (total circulation). OLS analogues show the same pattern. + $p < 0.10$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

Table A21: Subsample Headline IV by Region

Subsample	Coefficient (SE)	First-stage F	County-years
Northern Core (Non-South excl. KY/MD/MO/WV/DE/OK)	0.558*** (0.093)	261	9,355
Border South (KY/MD/MO/WV/DE)	0.112** (0.038)	624	1,872
Confederate 11 South	−0.109 (0.132)	109	6,628
<i>Memo:</i> Non-South (Confederate 11), headline	0.250*** (0.040)	1,000	11,689

Note: IV estimates of *radio* on presidential Democratic vote share, identical specification to the headline (Table 2, Panel A, column 2): county and state-by-year fixed effects; controls and instruments as there. Standard errors clustered at county level in parentheses. Oklahoma (N=462 county-years; ICPSR state code 53; admitted 1907; literacy test until 1916; partial white primary) is excluded from the Border row because of its mixed Jim Crow history; estimated separately it yields 0.141 (SE 0.056), in line with the Border row. * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

F Legislator Ideology

The district-level ideology estimates in Table 5 are read for their sign and bipartisan structure rather than their magnitudes. The two tables here show why: Table A22 reports the estimates under census-division-by-Congress fixed effects, where the within-division first stage is too thin to re-establish the magnitudes, and Table A23 isolates the within-legislator adaptation margin from selection.

The two parties responded to radio through different channels. Adding legislator fixed effects to the Nokken-Poole specification absorbs all between-legislator variation — including the replacement of incumbents by successors — so the radio coefficient identifies within-legislator adaptation alone (Table A23). Among high-illiteracy Democrats this estimate is $-\$0.61$ ($p < 0.05$): sitting incumbents moved their own roll-call behavior leftward, over and above being replaced by more liberal successors. Among high-illiteracy Republicans it is indistinguishable from zero, so the Republican response operated through selection alone. A pooled specification interacting radio with party confirms that the two adaptation estimates differ at conventional levels (interaction $p = 0.048$, with the same-signed difference weaker in the full sample, consistent with the effect concentrating where illiteracy was high), and the cross-measure comparison in Table 5 — where the Nokken-Poole estimate exceeds the DW-NOMINATE estimate in absolute value among Democrats but not Republicans — points the same way. Two features make these adaptation estimates conservative: legislator fixed effects are identified only off legislators serving two or more Congresses, and the period-specific Nokken-Poole score is noisier than the career-pooled DW-NOMINATE measure, so within-legislator differencing biases the coefficient toward zero — the Democratic estimate is significant despite this attenuation. Because the design uses Congress rather than state-by-Congress fixed effects, these estimates share the between-state identification caveat noted for the district-level results; the pooled interaction recovers the Democratic subsample estimate ($-\$0.59$ versus $-\$0.61$), confirming it as the correct analogue.

This channel asymmetry closes the two-party version of a signaling account, in which both leaderships broadcast leftward signals, producing a bipartisan response that direction alone cannot reject. Fitted to the Republican pattern, that account would have to posit that Republican elites signaled future challengers but not sitting incumbents, an unmotivated asymmetry. The electorate account derives the split from the direction of the shock instead: because the electorate moved leftward, following it was a low-cost move for Democratic incumbents — toward their own party's program — whereas for Republicans it meant adopting their opponents' positions, costly under their party label and futile, since the voters it would court were already defecting. Among Republicans the demand was met by replacement rather than conversion. The within-legislator estimates establish that the channels differ, not why; I read the brand-cost mechanism as the explanation most consistent with the evidence rather than a separately identified result.

Table A22: Legislator Ideology Under Census-Division $\times$ Congress Fixed Effects

	Full	High illit	Low illit	Dem, Hi illit	Rep, Hi illit
<i>Panel A: Nokken-Poole first dimension</i>					
Baseline (Congress FE)	−1.794*** (0.234)	−1.550*** (0.281)	−1.203 (0.944)	−1.033*** (0.263)	−0.286 (0.256)
Division $\times$ Congress FE	−0.470 (0.647)	−0.036 (2.561)	0.226 (0.885)	0.464 (1.270)	0.120 (0.912)
First-stage F (div $\times$ Cong)	155	5	126	13	27
<i>Panel B: Nominate first dimension</i>					
Baseline (Congress FE)	−1.536*** (0.208)	−1.412*** (0.247)	−0.320 (0.846)	−0.726*** (0.180)	−0.669** (0.244)
Division $\times$ Congress FE	−0.881 (0.607)	−1.272 (2.394)	0.498 (0.810)	0.495 (1.118)	−0.153 (0.594)
First-stage F (div $\times$ Cong)	155	5	125	13	27
N (div $\times$ Cong, Panel A)	4,304	2,140	2,140	1,559	532

Note: District-level IV estimates with district FE and either Congress FE (baseline rows, replicating Table 5) or census-division $\times$ Congress FE; SE clustered at the district level. Division $\times$ Congress FE absorb an additional 56–57% of the instruments' residual variance beyond Congress FE (state-by-Congress FE, which would absorb 79–84%, are infeasible because many states elect fewer than five representatives per Congress). In the full sample the first stage stays strong ($F = 155$) and the estimates stay negative but lose precision; in the high-illiteracy and party subsamples the first stage collapses (F between 5 and 27), so those cells are uninformative rather than contradictory. This confirms the caveat in the text: district-level identification rests largely on between-state variation, which is why the text reads Table 5 for its sign pattern and bipartisan structure rather than its magnitudes. $^+p < 0.10$, $^*p < 0.05$, $^{**}p < 0.01$, $^{***}p < 0.001$.

Table A23: Within-Legislator Adaptation: District versus Legislator Fixed Effects

	Full	High illit	Low illit	Dem, Hi illit	Rep, Hi illit
<i>Panel A: Nokken-Poole first dimension (adaptation channel)</i>					
District FE (adaptation + selection)	−1.794*** (0.234)	−1.550*** (0.281)	−1.203 (0.944)	−1.033*** (0.263)	−0.286 (0.256)
Legislator FE (adaptation only)	−0.450* (0.186)	−0.422* (0.205)	−1.651* (0.805)	−0.612* (0.289)	0.573 (0.413)
First-stage F (legislator FE)	379	233	31	129	80
<i>Panel B: DW-NOMINATE first dimension (selection channel)</i>					
District FE (selection only)	−1.536*** (0.208)	−1.412*** (0.247)	−0.320 (0.846)	−0.726*** (0.180)	−0.669** (0.244)
Legislator FE (career-static; absorbed)	0.000	0.000	0.000	0.000	0.000
N (legislator FE, Panel A)	3,861	1,959	1,900	1,469	485

Note: District-level IV estimates; the endogenous radio term is instrumented by ground conductivity $\times$ time and woodland $\times$ time. District-FE rows use district and Congress fixed effects (reproducing Table 5); legislator-FE rows replace district with legislator FE, identifying the within-legislator adaptation margin. SE (in parentheses) clustered at the district level (district-FE rows) or legislator level (legislator-FE rows). Because common-space DW-NOMINATE is career-static, it is absorbed entirely by legislator FE (Panel B legislator-FE row $\equiv$ 0), confirming the specification isolates adaptation rather than selection. Legislator FE are identified off the 67% of legislators serving two or more Congresses; the Low-illiteracy cell has a weak first stage ($F = 31$) and should not be read. *Formal asymmetry test:* a pooled within-legislator IV interacting radio with party (high-illiteracy sample) gives a Democratic adaptation of -0.59 ($p < 0.05$), a Republican – Democratic gap of $+0.85$ ($p = 0.048$), and a Republican adaptation indistinguishable from zero ($+0.26$); the gap is same-signed but not significant in the full sample. $^+p < 0.10$, $^*p < 0.05$, $^{**}p < 0.01$, $^{***}p < 0.001$.